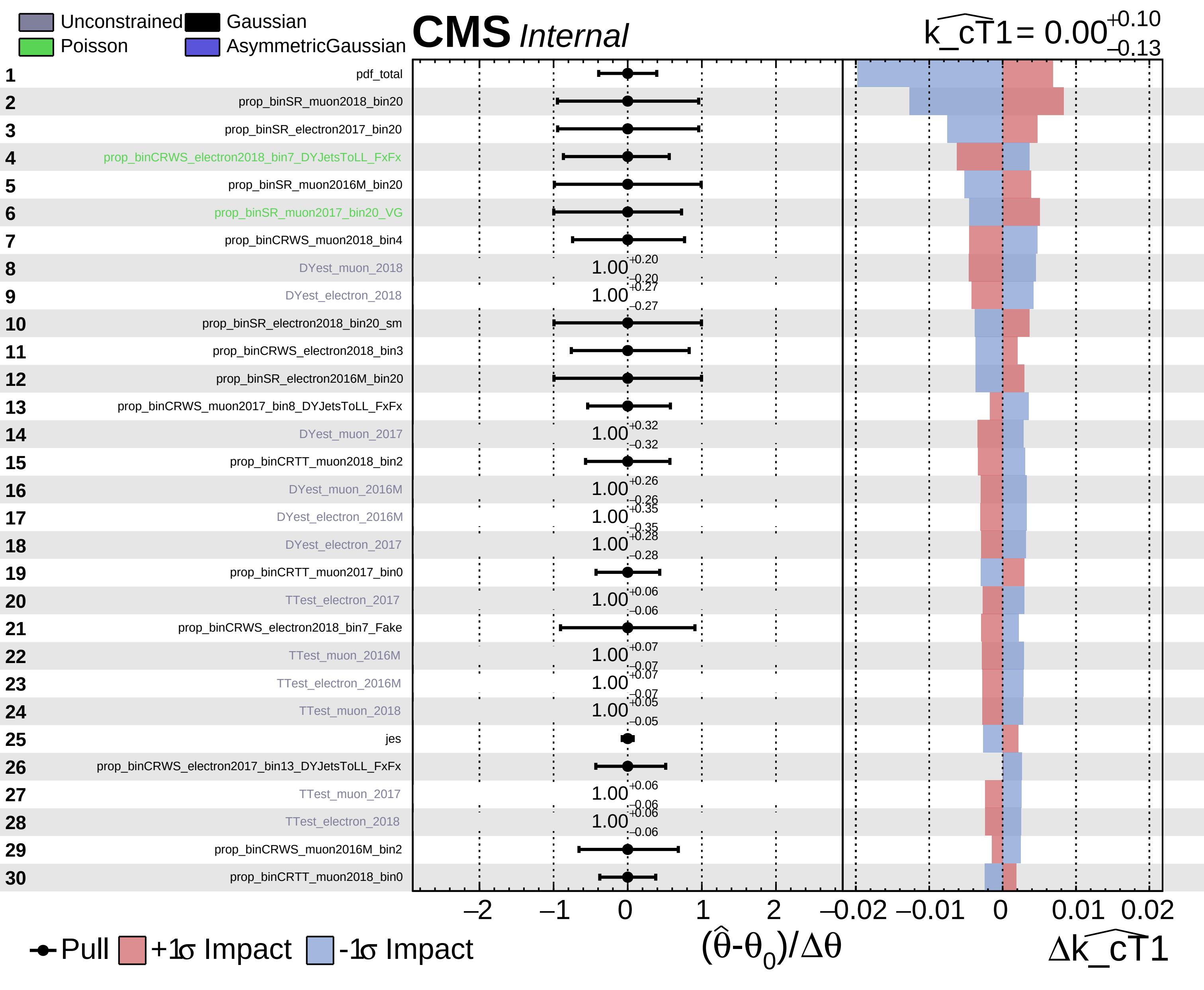
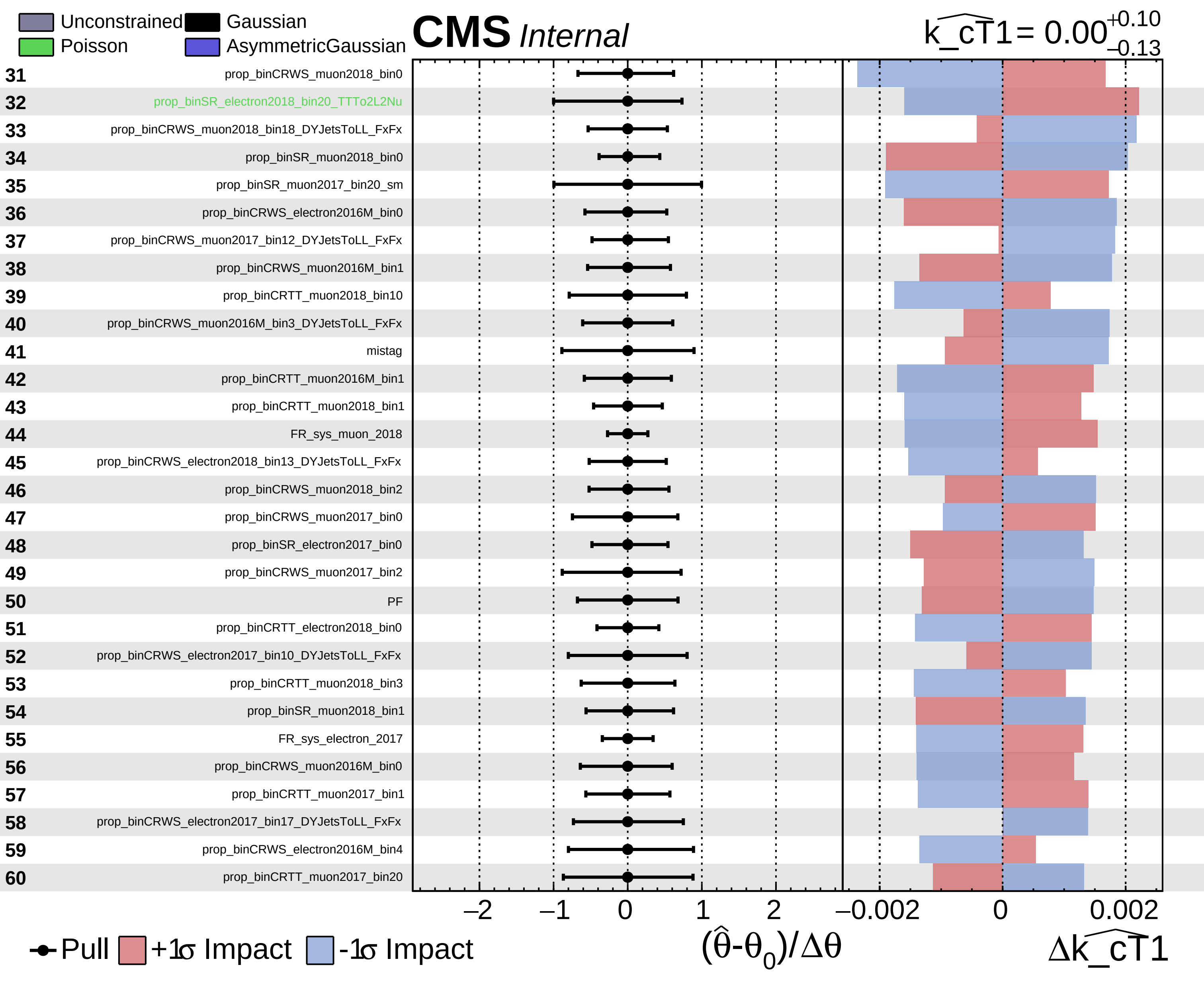
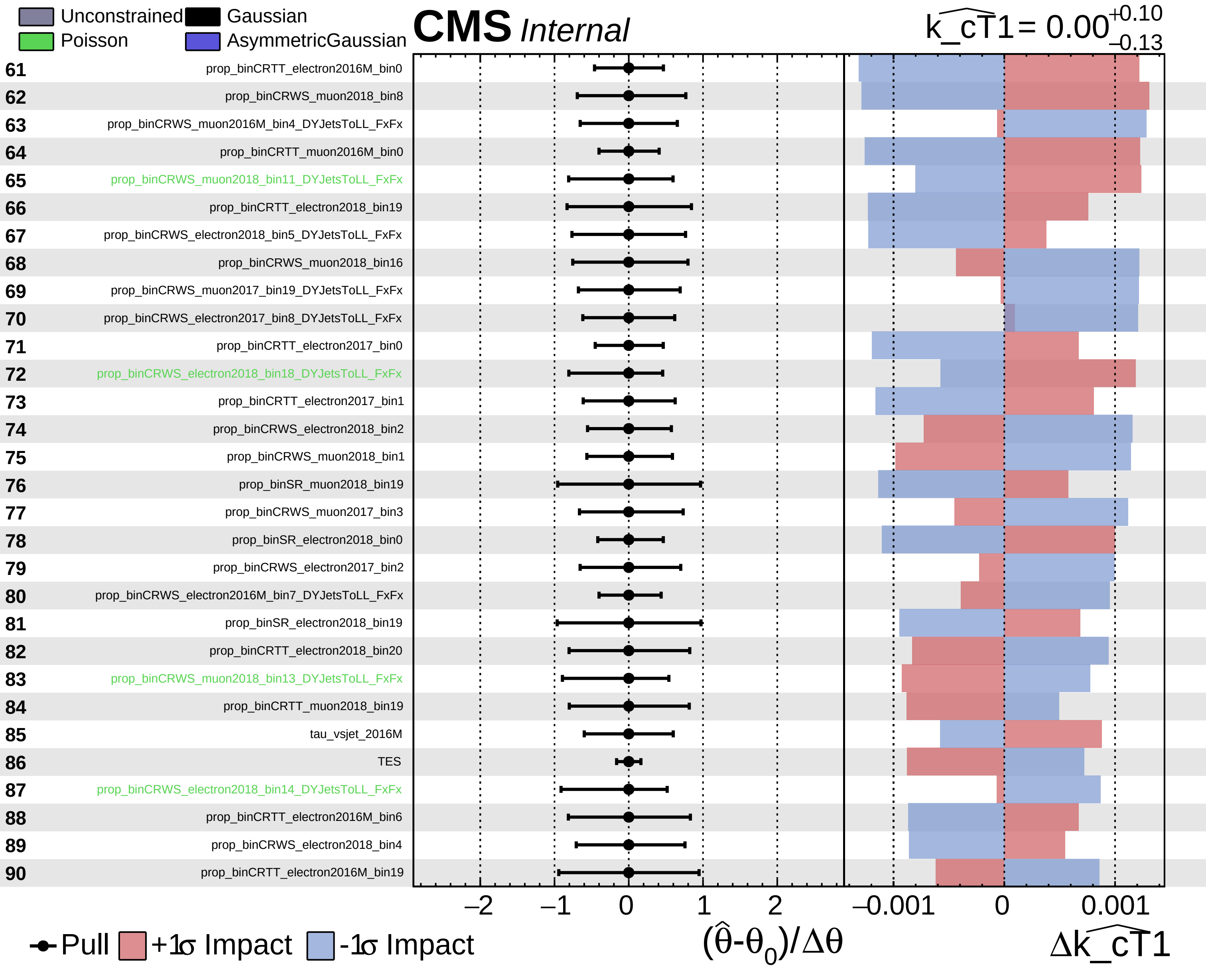


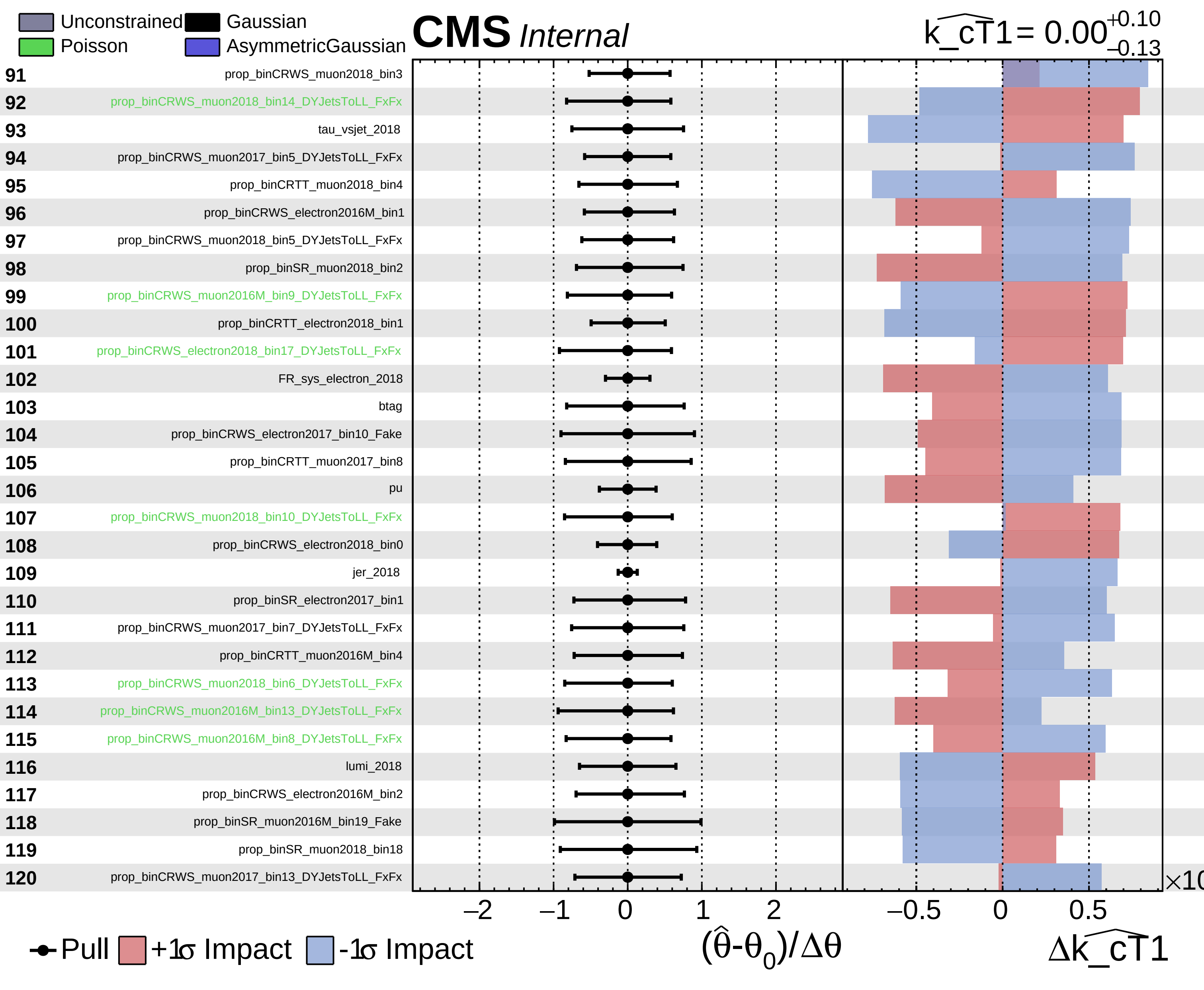
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
1 pdf_total
2 prop_binSR_muon2018_bin20
3 prop_binSR_electron2017_bin20
4 prop_binCRWS_electron2018_bin7_DYJetsToLL_FxFx
5 prop_binSR_muon2016M_bin20
6 prop_binSR_muon2017_bin20_VG
7 prop_binCRWS_muon2018_bin4
8 DYest_muon_2018 1.00 +0.20 −0.20
9 DYest_electron_2018 1.00 +0.27 −0.27
10 prop_binSR_electron2018_bin20_sm
11 prop_binCRWS_electron2018_bin3
12 prop_binSR_electron2016M_bin20
13 prop_binCRWS_muon2017_bin8_DYJetsToLL_FxFx
14 DYest_muon_2017 1.00 +0.32 −0.32
15 prop_binCRTT_muon2018_bin2
16 DYest_muon_2016M 1.00 +0.26 −0.26
17 DYest_electron_2016M 1.00 +0.35 −0.35
18 DYest_electron_2017 1.00 +0.28 −0.28
19 prop_binCRTT_muon2017_bin0
20 TTest_electron_2017 1.00 +0.06 −0.06
21 prop_binCRWS_electron2018_bin7_Fake
22 TTest_muon_2016M 1.00 +0.07 −0.07
23 TTest_electron_2016M 1.00 +0.07 −0.07
24 TTest_muon_2018 1.00 +0.05 −0.05
25 jes
26 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
27 TTest_muon_2017 1.00 +0.06 −0.06
28 TTest_electron_2018 1.00 +0.06 −0.06
29 prop_binCRWS_muon2016M_bin2
30 prop_binCRTT_muon2018_bin0
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



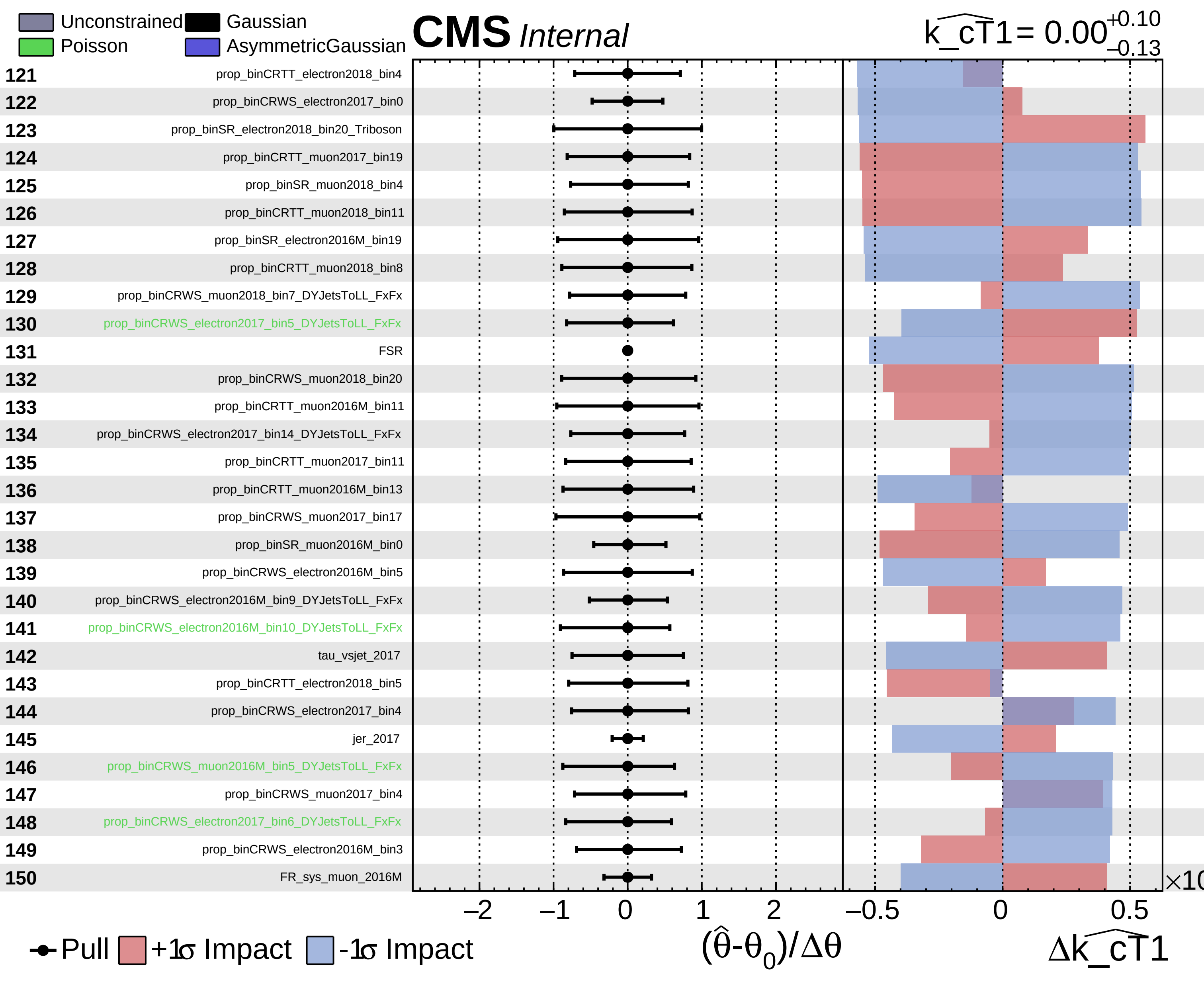
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
31 prop_binCRWS_muon2018_bin0
32 prop_binSR_electron2018_bin20_TTTo2L2Nu
33 prop_binCRWS_muon2018_bin18_DYJetsToLL_FxFx
34 prop_binSR_muon2018_bin0
35 prop_binSR_muon2017_bin20_sm
36 prop_binCRWS_electron2016M_bin0
37 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
38 prop_binCRWS_muon2016M_bin1
39 prop_binCRTT_muon2018_bin10
40 prop_binCRWS_muon2016M_bin3_DYJetsToLL_FxFx
41 mistag
42 prop_binCRTT_muon2016M_bin1
43 prop_binCRTT_muon2018_bin1
44 FR_sys_muon_2018
45 prop_binCRWS_electron2018_bin13_DYJetsToLL_FxFx
46 prop_binCRWS_muon2018_bin2
47 prop_binCRWS_muon2017_bin0
48 prop_binSR_electron2017_bin0
49 prop_binCRWS_muon2017_bin2
50 PF
51 prop_binCRTT_electron2018_bin0
52 prop_binCRWS_electron2017_bin10_DYJetsToLL_FxFx
53 prop_binCRTT_muon2018_bin3
54 prop_binSR_muon2018_bin1
55 FR_sys_electron_2017
56 prop_binCRWS_muon2016M_bin0
57 prop_binCRTT_muon2017_bin1
58 prop_binCRWS_electron2017_bin17_DYJetsToLL_FxFx
59 prop_binCRWS_electron2016M_bin4
60 prop_binCRTT_muon2017_bin20
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



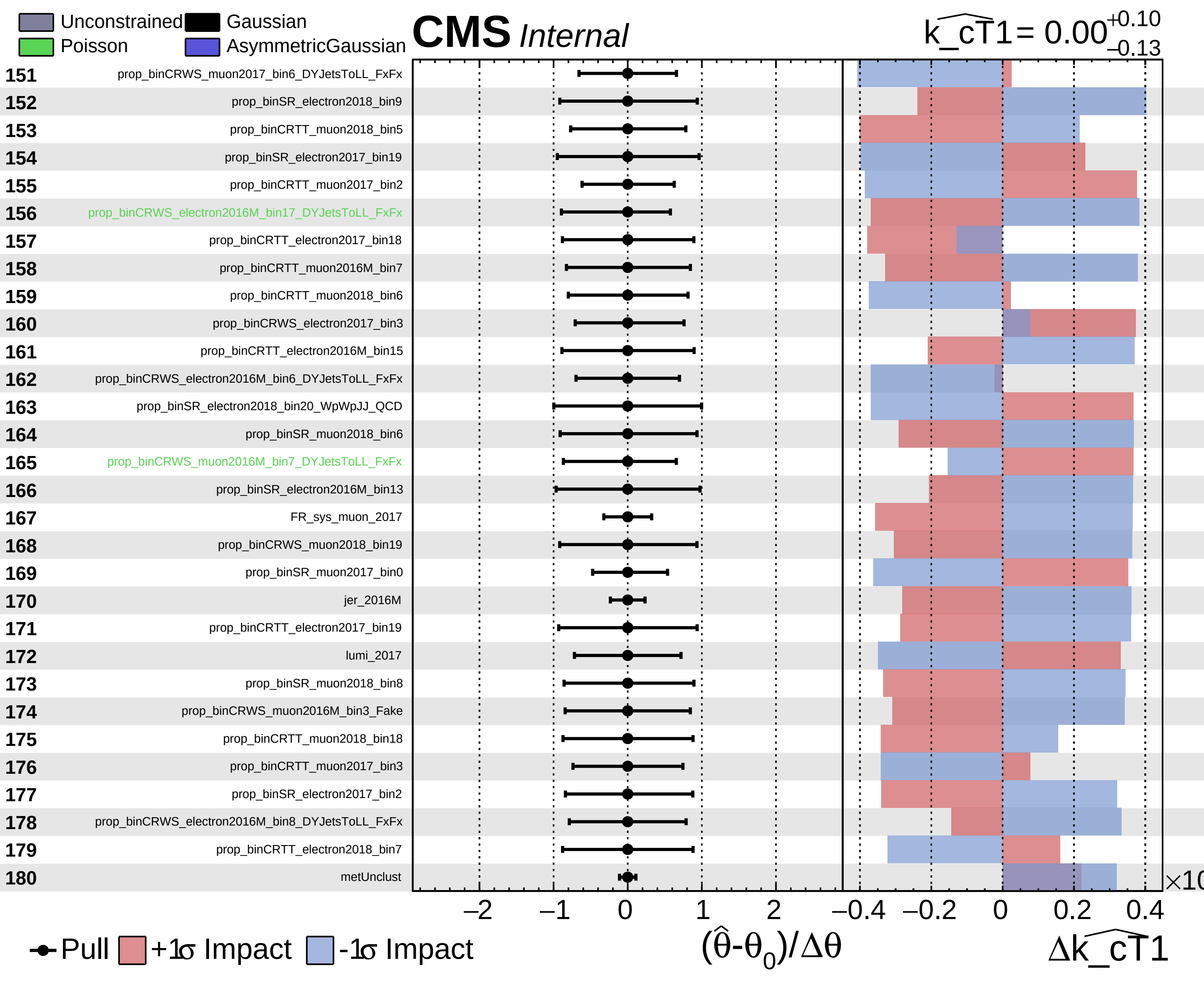
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
61 prop_binCRTT_electron2016M_bin0
62 prop_binCRWS_muon2018_bin8
63 prop_binCRWS_muon2016M_bin4_DYJetsToLL_FxFx
64 prop_binCRTT_muon2016M_bin0
65 prop_binCRWS_muon2018_bin11_DYJetsToLL_FxFx
66 prop_binCRTT_electron2018_bin19
67 prop_binCRWS_electron2018_bin5_DYJetsToLL_FxFx
68 prop_binCRWS_muon2018_bin16
69 prop_binCRWS_muon2017_bin19_DYJetsToLL_FxFx
70 prop_binCRWS_electron2017_bin8_DYJetsToLL_FxFx
71 prop_binCRTT_electron2017_bin0
72 prop_binCRWS_electron2018_bin18_DYJetsToLL_FxFx
73 prop_binCRTT_electron2017_bin1
74 prop_binCRWS_electron2018_bin2
75 prop_binCRWS_muon2018_bin1
76 prop_binSR_muon2018_bin19
77 prop_binCRWS_muon2017_bin3
78 prop_binSR_electron2018_bin0
79 prop_binCRWS_electron2017_bin2
80 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
81 prop_binSR_electron2018_bin19
82 prop_binCRTT_electron2018_bin20
83 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
84 prop_binCRTT_muon2018_bin19
85 tau_vsjet_2016M
86 TES
87 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
88 prop_binCRTT_electron2016M_bin6
89 prop_binCRWS_electron2018_bin4
90 prop_binCRTT_electron2016M_bin19
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



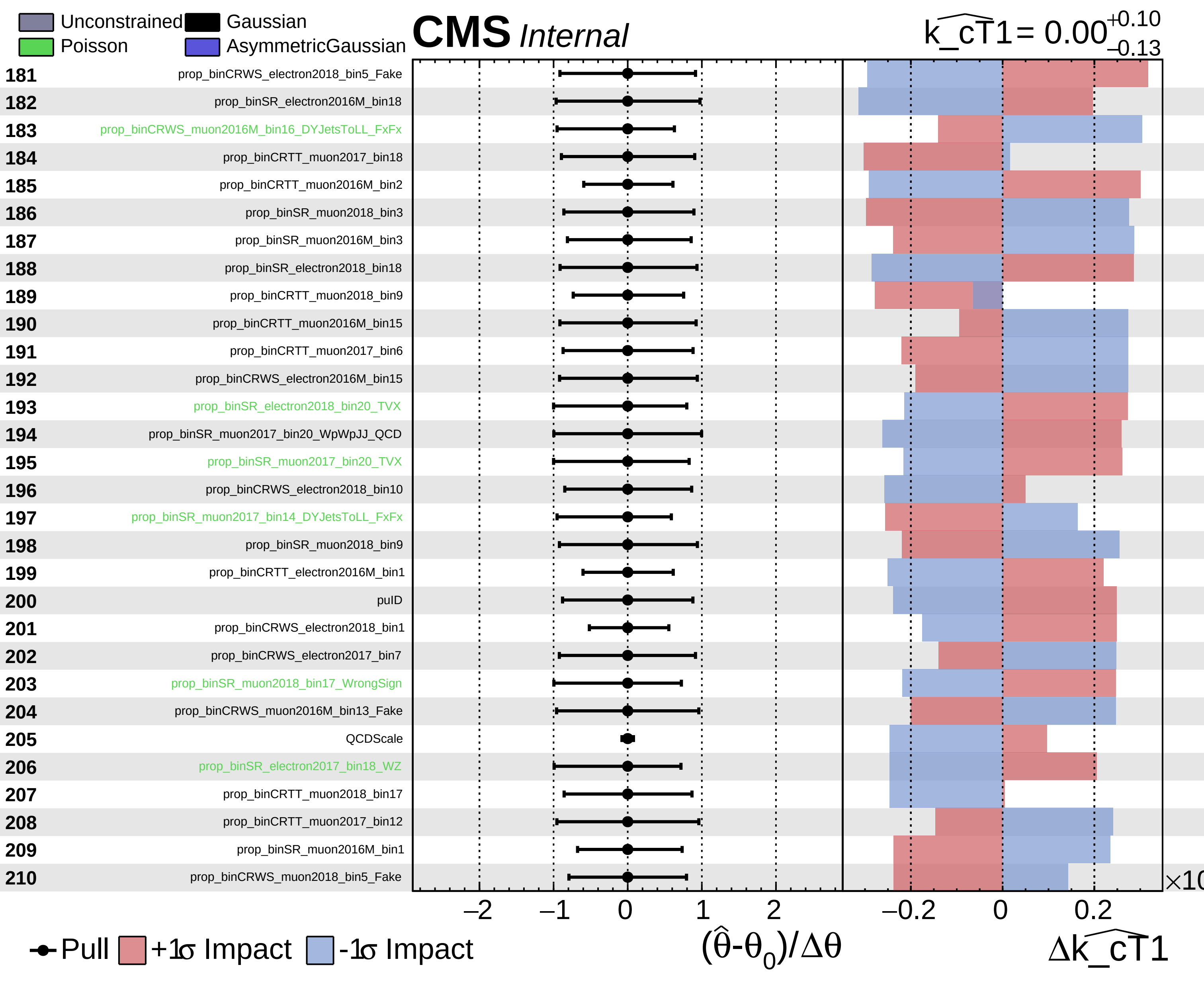
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
91 prop_binCRWS_muon2018_bin3
92 prop_binCRWS_muon2018_bin14_DYJetsToLL_FxFx
93 tau_vsjet_2018
94 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
95 prop_binCRTT_muon2018_bin4
96 prop_binCRWS_electron2016M_bin1
97 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
98 prop_binSR_muon2018_bin2
99 prop_binCRWS_muon2016M_bin9_DYJetsToLL_FxFx
100 prop_binCRTT_electron2018_bin1
101 prop_binCRWS_electron2018_bin17_DYJetsToLL_FxFx
102 FR_sys_electron_2018
103 btag
104 prop_binCRWS_electron2017_bin10_Fake
105 prop_binCRTT_muon2017_bin8
106 pu
107 prop_binCRWS_muon2018_bin10_DYJetsToLL_FxFx
108 prop_binCRWS_electron2018_bin0
109 jer_2018
110 prop_binSR_electron2017_bin1
111 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
112 prop_binCRTT_muon2016M_bin4
113 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
114 prop_binCRWS_muon2016M_bin13_DYJetsToLL_FxFx
115 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
116 lumi_2018
117 prop_binCRWS_electron2016M_bin2
118 prop_binSR_muon2016M_bin19_Fake
119 prop_binSR_muon2018_bin18
120 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



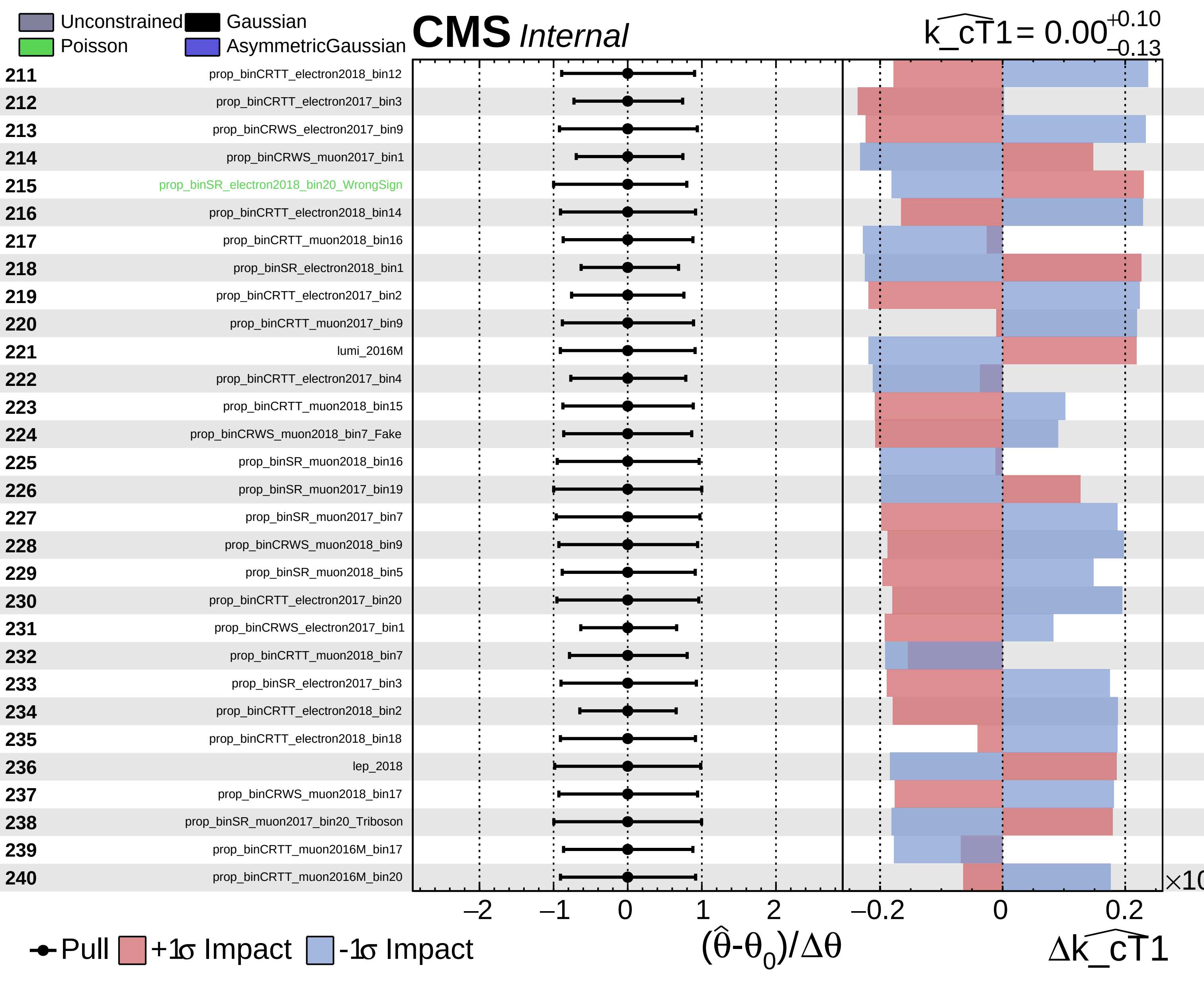
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
121 prop_binCRTT_electron2018_bin4
122 prop_binCRWS_electron2017_bin0
123 prop_binSR_electron2018_bin20_Triboson
124 prop_binCRTT_muon2017_bin19
125 prop_binSR_muon2018_bin4
126 prop_binCRTT_muon2018_bin11
127 prop_binSR_electron2016M_bin19
128 prop_binCRTT_muon2018_bin8
129 prop_binCRWS_muon2018_bin7_DYJetsToLL_FxFx
130 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
131 FSR
132 prop_binCRWS_muon2018_bin20
133 prop_binCRTT_muon2016M_bin11
134 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
135 prop_binCRTT_muon2017_bin11
136 prop_binCRTT_muon2016M_bin13
137 prop_binCRWS_muon2017_bin17
138 prop_binSR_muon2016M_bin0
139 prop_binCRWS_electron2016M_bin5
140 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
141 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
142 tau_vsjet_2017
143 prop_binCRTT_electron2018_bin5
144 prop_binCRWS_electron2017_bin4
145 jer_2017
146 prop_binCRWS_muon2016M_bin5_DYJetsToLL_FxFx
147 prop_binCRWS_muon2017_bin4
148 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
149 prop_binCRWS_electron2016M_bin3
150 FR_sys_muon_2016M
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



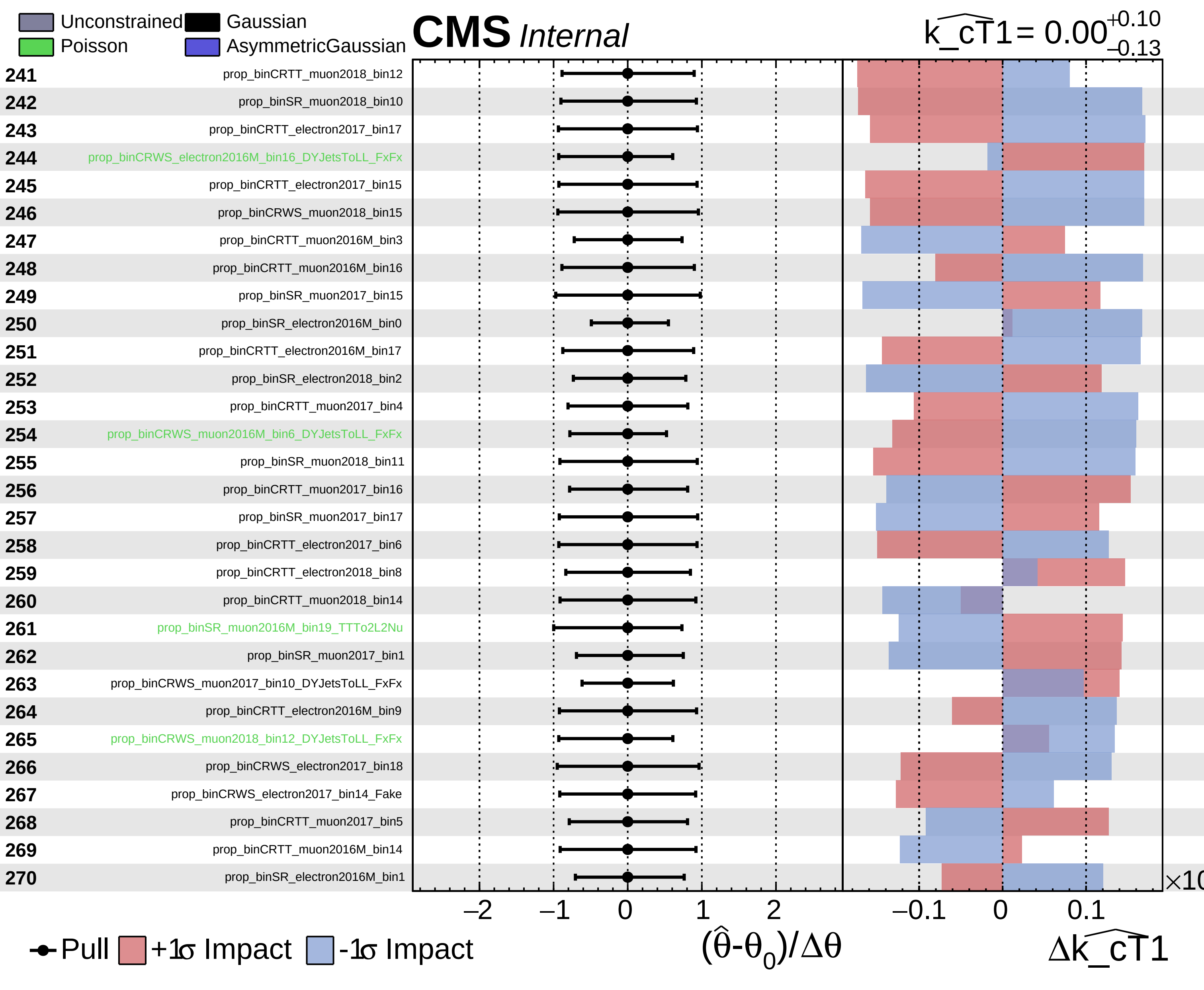
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
151 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
152 prop_binSR_electron2018_bin9
153 prop_binCRTT_muon2018_bin5
154 prop_binSR_electron2017_bin19
155 prop_binCRTT_muon2017_bin2
156 prop_binCRWS_electron2016M_bin17_DYJetsToLL_FxFx
157 prop_binCRTT_electron2017_bin18
158 prop_binCRTT_muon2016M_bin7
159 prop_binCRTT_muon2018_bin6
160 prop_binCRWS_electron2017_bin3
161 prop_binCRTT_electron2016M_bin15
162 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
163 prop_binSR_electron2018_bin20_WpWpJJ_QCD
164 prop_binSR_muon2018_bin6
165 prop_binCRWS_muon2016M_bin7_DYJetsToLL_FxFx
166 prop_binSR_electron2016M_bin13
167 FR_sys_muon_2017
168 prop_binCRWS_muon2018_bin19
169 prop_binSR_muon2017_bin0
170 jer_2016M
171 prop_binCRTT_electron2017_bin19
172 lumi_2017
173 prop_binSR_muon2018_bin8
174 prop_binCRWS_muon2016M_bin3_Fake
175 prop_binCRTT_muon2018_bin18
176 prop_binCRTT_muon2017_bin3
177 prop_binSR_electron2017_bin2
178 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
179 prop_binCRTT_electron2018_bin7
180 metUnclust
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



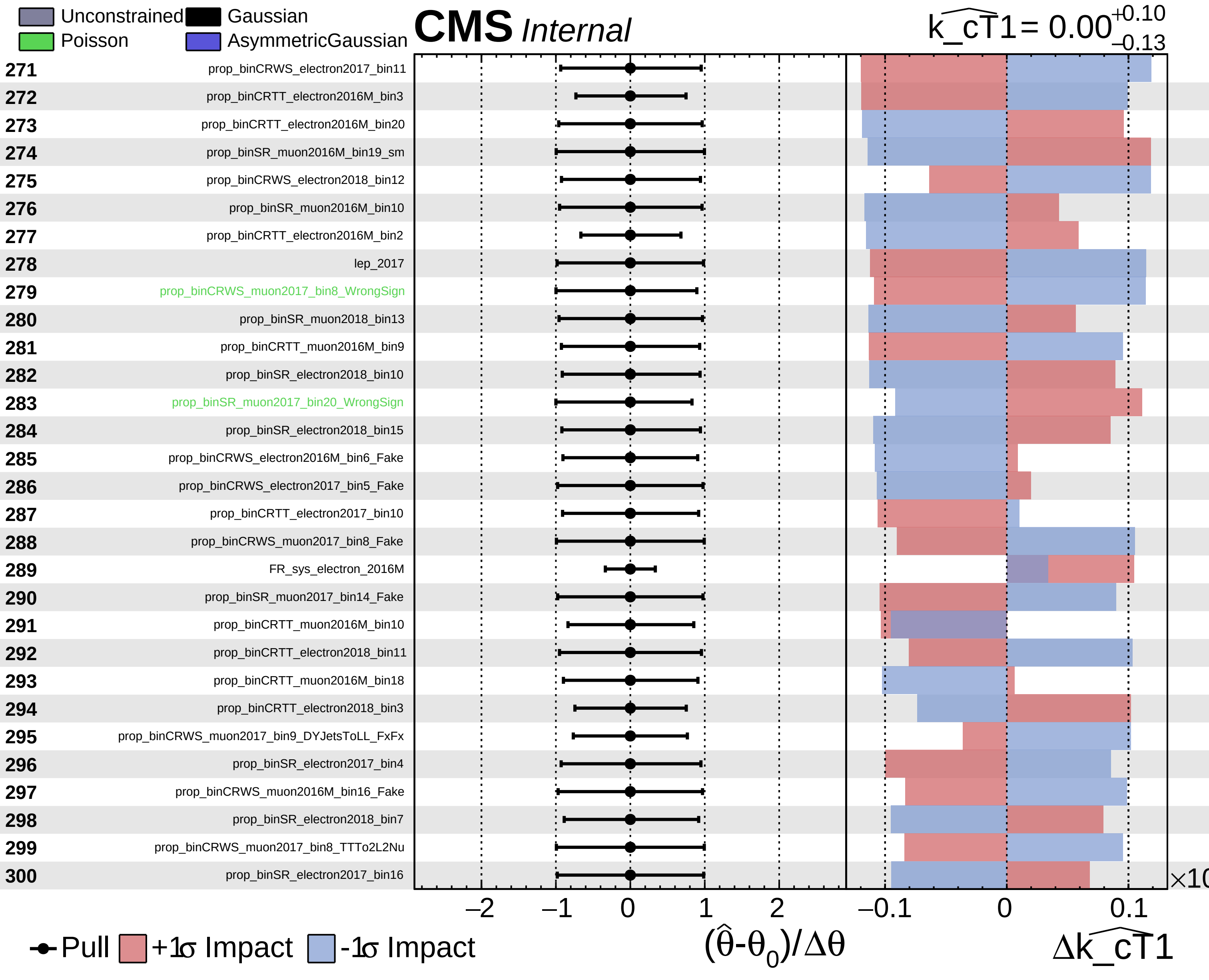
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
181 prop_binCRWS_electron2018_bin5_Fake
182 prop_binSR_electron2016M_bin18
183 prop_binCRWS_muon2016M_bin16_DYJetsToLL_FxFx
184 prop_binCRTT_muon2017_bin18
185 prop_binCRTT_muon2016M_bin2
186 prop_binSR_muon2018_bin3
187 prop_binSR_muon2016M_bin3
188 prop_binSR_electron2018_bin18
189 prop_binCRTT_muon2018_bin9
190 prop_binCRTT_muon2016M_bin15
191 prop_binCRTT_muon2017_bin6
192 prop_binCRWS_electron2016M_bin15
193 prop_binSR_electron2018_bin20_TVX
194 prop_binSR_muon2017_bin20_WpWpJJ_QCD
195 prop_binSR_muon2017_bin20_TVX
196 prop_binCRWS_electron2018_bin10
197 prop_binSR_muon2017_bin14_DYJetsToLL_FxFx
198 prop_binSR_muon2018_bin9
199 prop_binCRTT_electron2016M_bin1
200 puID
201 prop_binCRWS_electron2018_bin1
202 prop_binCRWS_electron2017_bin7
203 prop_binSR_muon2018_bin17_WrongSign
204 prop_binCRWS_muon2016M_bin13_Fake
205 QCDScale
206 prop_binSR_electron2017_bin18_WZ
207 prop_binCRTT_muon2018_bin17
208 prop_binCRTT_muon2017_bin12
209 prop_binSR_muon2016M_bin1
210 prop_binCRWS_muon2018_bin5_Fake
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



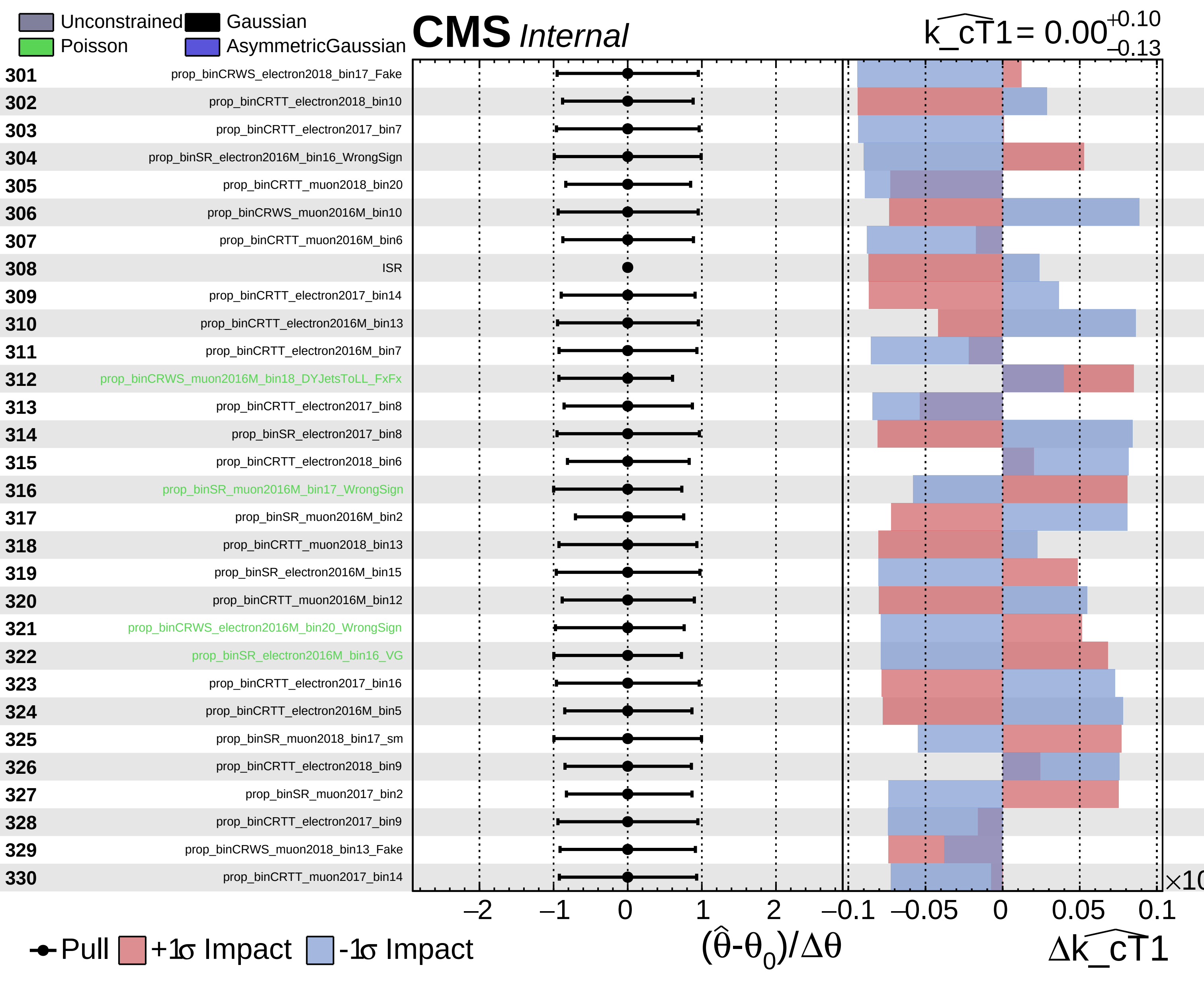
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
211 prop_binCRTT_electron2018_bin12
212 prop_binCRTT_electron2017_bin3
213 prop_binCRWS_electron2017_bin9
214 prop_binCRWS_muon2017_bin1
215 prop_binSR_electron2018_bin20_WrongSign
216 prop_binCRTT_electron2018_bin14
217 prop_binCRTT_muon2018_bin16
218 prop_binSR_electron2018_bin1
219 prop_binCRTT_electron2017_bin2
220 prop_binCRTT_muon2017_bin9
221 lumi_2016M
222 prop_binCRTT_electron2017_bin4
223 prop_binCRTT_muon2018_bin15
224 prop_binCRWS_muon2018_bin7_Fake
225 prop_binSR_muon2018_bin16
226 prop_binSR_muon2017_bin19
227 prop_binSR_muon2017_bin7
228 prop_binCRWS_muon2018_bin9
229 prop_binSR_muon2018_bin5
230 prop_binCRTT_electron2017_bin20
231 prop_binCRWS_electron2017_bin1
232 prop_binCRTT_muon2018_bin7
233 prop_binSR_electron2017_bin3
234 prop_binCRTT_electron2018_bin2
235 prop_binCRTT_electron2018_bin18
236 lep_2018
237 prop_binCRWS_muon2018_bin17
238 prop_binSR_muon2017_bin20_Triboson
239 prop_binCRTT_muon2016M_bin17
240 prop_binCRTT_muon2016M_bin20
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



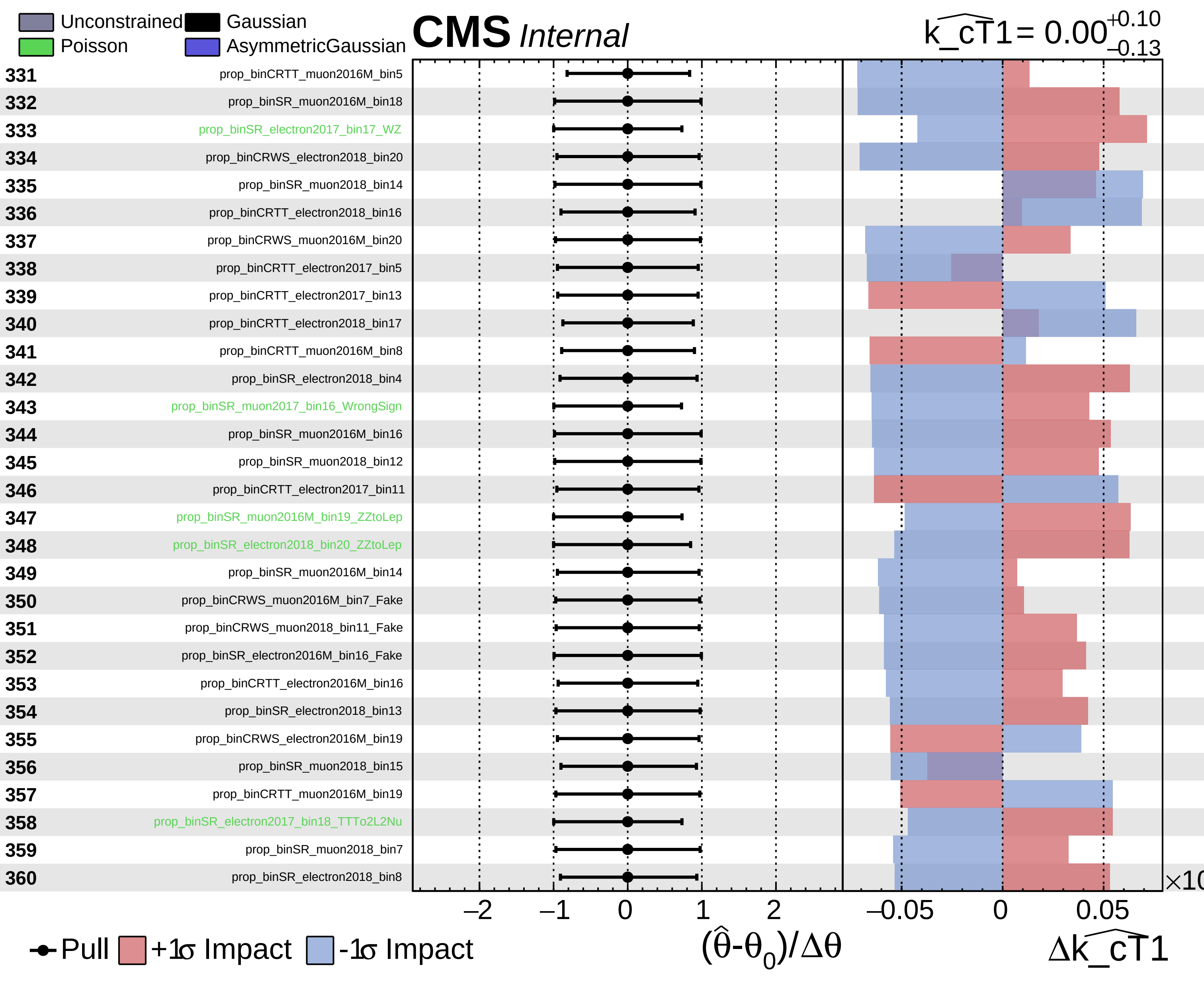
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
241 prop_binCRTT_muon2018_bin12
242 prop_binSR_muon2018_bin10
243 prop_binCRTT_electron2017_bin17
244 prop_binCRWS_electron2016M_bin16_DYJetsToLL_FxFx
245 prop_binCRTT_electron2017_bin15
246 prop_binCRWS_muon2018_bin15
247 prop_binCRTT_muon2016M_bin3
248 prop_binCRTT_muon2016M_bin16
249 prop_binSR_muon2017_bin15
250 prop_binSR_electron2016M_bin0
251 prop_binCRTT_electron2016M_bin17
252 prop_binSR_electron2018_bin2
253 prop_binCRTT_muon2017_bin4
254 prop_binCRWS_muon2016M_bin6_DYJetsToLL_FxFx
255 prop_binSR_muon2018_bin11
256 prop_binCRTT_muon2017_bin16
257 prop_binSR_muon2017_bin17
258 prop_binCRTT_electron2017_bin6
259 prop_binCRTT_electron2018_bin8
260 prop_binCRTT_muon2018_bin14
261 prop_binSR_muon2016M_bin19_TTTo2L2Nu
262 prop_binSR_muon2017_bin1
263 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
264 prop_binCRTT_electron2016M_bin9
265 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
266 prop_binCRWS_electron2017_bin18
267 prop_binCRWS_electron2017_bin14_Fake
268 prop_binCRTT_muon2017_bin5
269 prop_binCRTT_muon2016M_bin14
270 prop_binSR_electron2016M_bin1
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



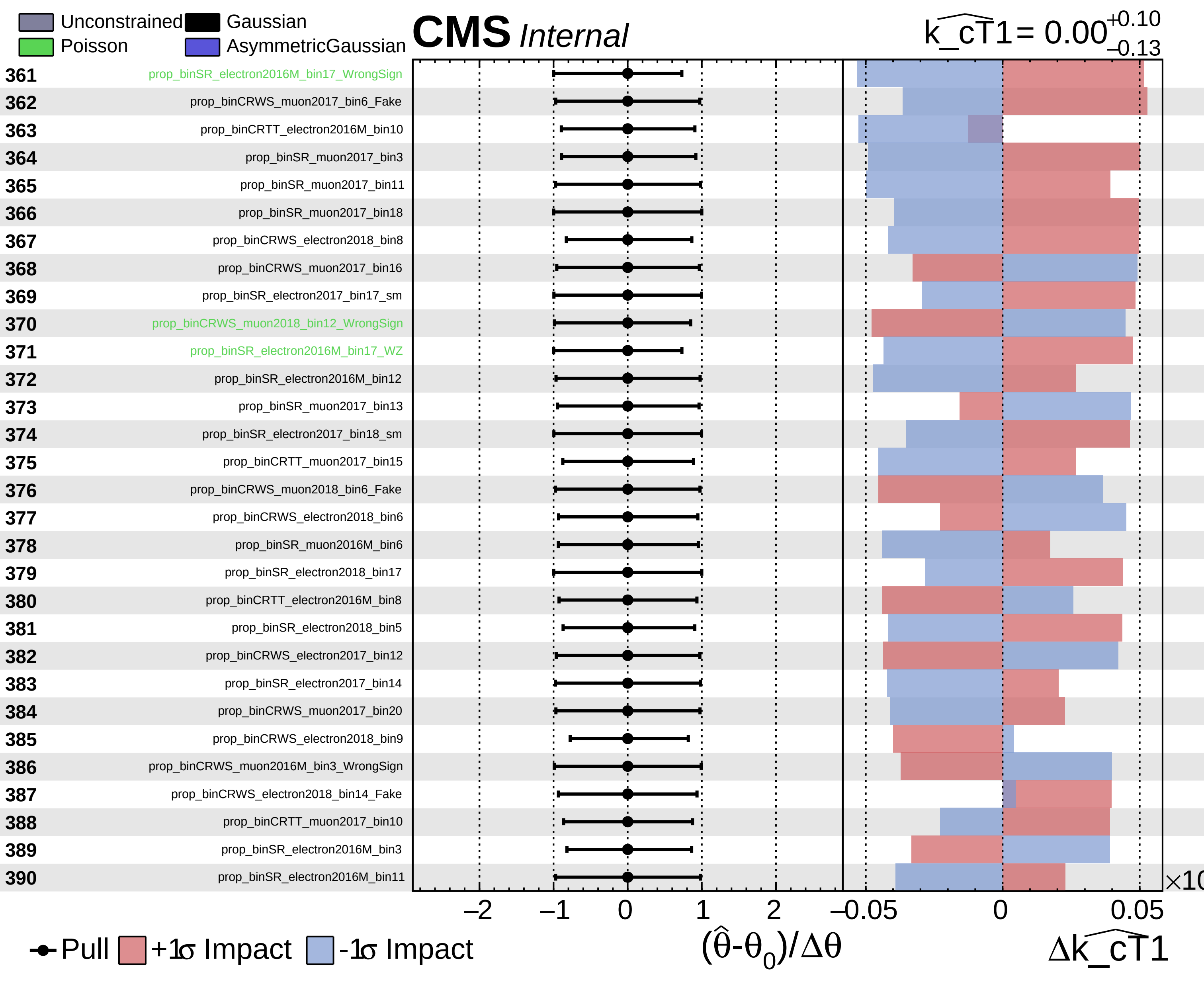
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
271 prop_binCRWS_electron2017_bin11
272 prop_binCRTT_electron2016M_bin3
273 prop_binCRTT_electron2016M_bin20
274 prop_binSR_muon2016M_bin19_sm
275 prop_binCRWS_electron2018_bin12
276 prop_binSR_muon2016M_bin10
277 prop_binCRTT_electron2016M_bin2
278 lep_2017
279 prop_binCRWS_muon2017_bin8_WrongSign
280 prop_binSR_muon2018_bin13
281 prop_binCRTT_muon2016M_bin9
282 prop_binSR_electron2018_bin10
283 prop_binSR_muon2017_bin20_WrongSign
284 prop_binSR_electron2018_bin15
285 prop_binCRWS_electron2016M_bin6_Fake
286 prop_binCRWS_electron2017_bin5_Fake
287 prop_binCRTT_electron2017_bin10
288 prop_binCRWS_muon2017_bin8_Fake
289 FR_sys_electron_2016M
290 prop_binSR_muon2017_bin14_Fake
291 prop_binCRTT_muon2016M_bin10
292 prop_binCRTT_electron2018_bin11
293 prop_binCRTT_muon2016M_bin18
294 prop_binCRTT_electron2018_bin3
295 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
296 prop_binSR_electron2017_bin4
297 prop_binCRWS_muon2016M_bin16_Fake
298 prop_binSR_electron2018_bin7
299 prop_binCRWS_muon2017_bin8_TTTo2L2Nu
300 prop_binSR_electron2017_bin16
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



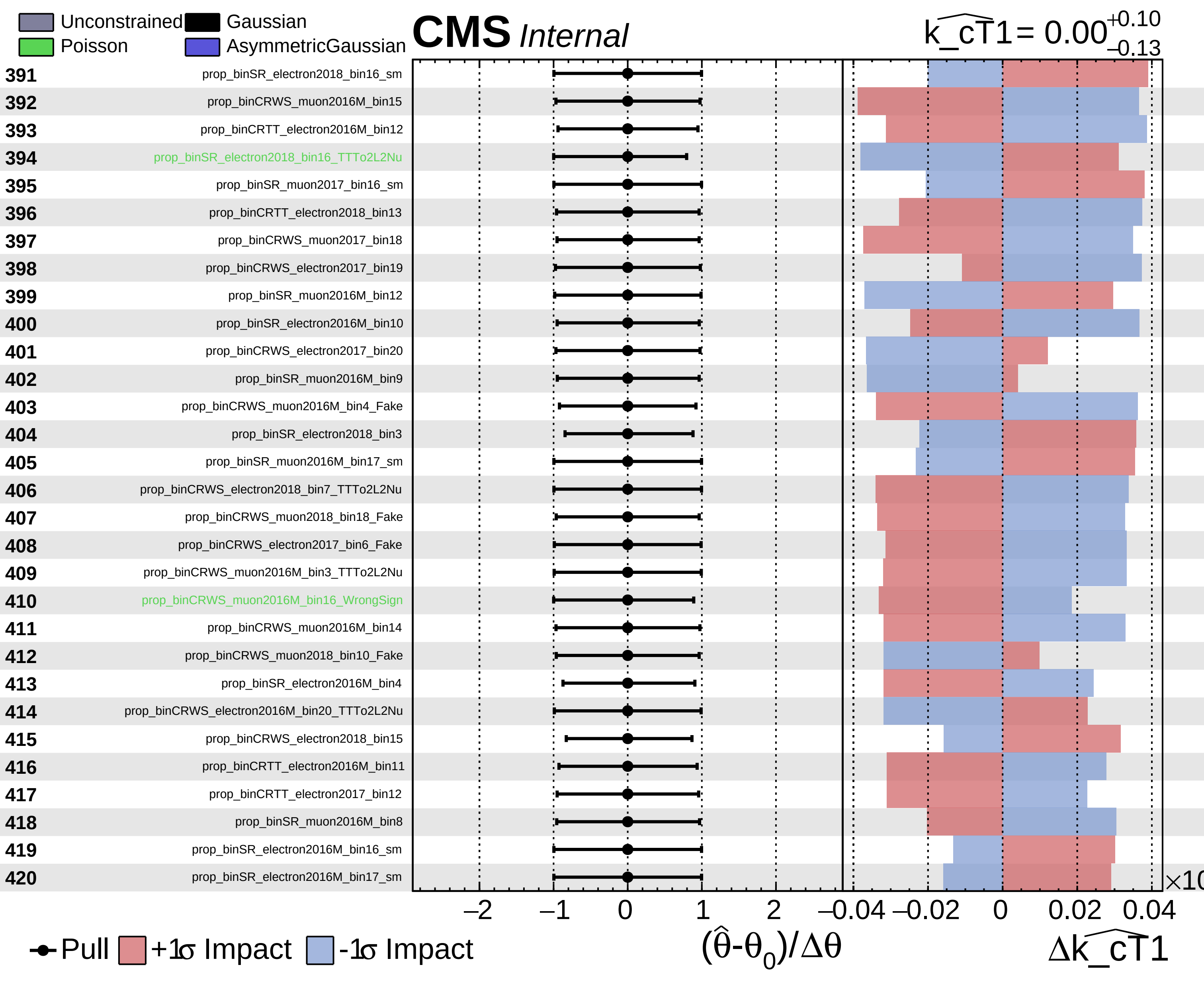
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
301 prop_binCRWS_electron2018_bin17_Fake
302 prop_binCRTT_electron2018_bin10
303 prop_binCRTT_electron2017_bin7
304 prop_binSR_electron2016M_bin16_WrongSign
305 prop_binCRTT_muon2018_bin20
306 prop_binCRWS_muon2016M_bin10
307 prop_binCRTT_muon2016M_bin6
308 ISR
309 prop_binCRTT_electron2017_bin14
310 prop_binCRTT_electron2016M_bin13
311 prop_binCRTT_electron2016M_bin7
312 prop_binCRWS_muon2016M_bin18_DYJetsToLL_FxFx
313 prop_binCRTT_electron2017_bin8
314 prop_binSR_electron2017_bin8
315 prop_binCRTT_electron2018_bin6
316 prop_binSR_muon2016M_bin17_WrongSign
317 prop_binSR_muon2016M_bin2
318 prop_binCRTT_muon2018_bin13
319 prop_binSR_electron2016M_bin15
320 prop_binCRTT_muon2016M_bin12
321 prop_binCRWS_electron2016M_bin20_WrongSign
322 prop_binSR_electron2016M_bin16_VG
323 prop_binCRTT_electron2017_bin16
324 prop_binCRTT_electron2016M_bin5
325 prop_binSR_muon2018_bin17_sm
326 prop_binCRTT_electron2018_bin9
327 prop_binSR_muon2017_bin2
328 prop_binCRTT_electron2017_bin9
329 prop_binCRWS_muon2018_bin13_Fake
330 prop_binCRTT_muon2017_bin14
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta \widehat{k\_cT1}$

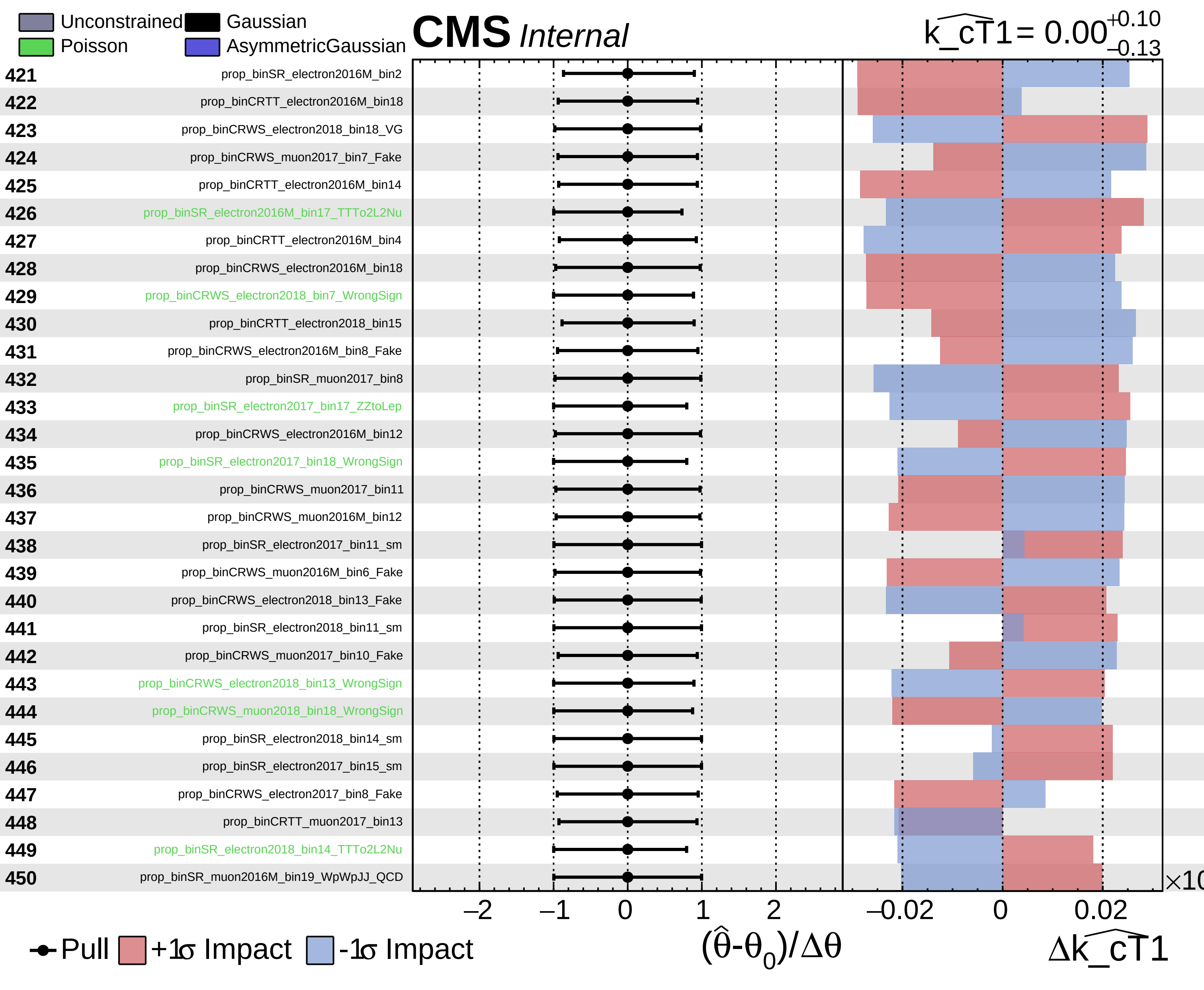


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
331 prop_binCRTT_muon2016M_bin5
332 prop_binSR_muon2016M_bin18
333 prop_binSR_electron2017_bin17_WZ
334 prop_binCRWS_electron2018_bin20
335 prop_binSR_muon2018_bin14
336 prop_binCRTT_electron2018_bin16
337 prop_binCRWS_muon2016M_bin20
338 prop_binCRTT_electron2017_bin5
339 prop_binCRTT_electron2017_bin13
340 prop_binCRTT_electron2018_bin17
341 prop_binCRTT_muon2016M_bin8
342 prop_binSR_electron2018_bin4
343 prop_binSR_muon2017_bin16_WrongSign
344 prop_binSR_muon2016M_bin16
345 prop_binSR_muon2018_bin12
346 prop_binCRTT_electron2017_bin11
347 prop_binSR_muon2016M_bin19_ZZtoLep
348 prop_binSR_electron2018_bin20_ZZtoLep
349 prop_binSR_muon2016M_bin14
350 prop_binCRWS_muon2016M_bin7_Fake
351 prop_binCRWS_muon2018_bin11_Fake
352 prop_binSR_electron2016M_bin16_Fake
353 prop_binCRTT_electron2016M_bin16
354 prop_binSR_electron2018_bin13
355 prop_binCRWS_electron2016M_bin19
356 prop_binSR_muon2018_bin15
357 prop_binCRTT_muon2016M_bin19
358 prop_binSR_electron2017_bin18_TTTo2L2Nu
359 prop_binSR_muon2018_bin7
360 prop_binSR_electron2018_bin8
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

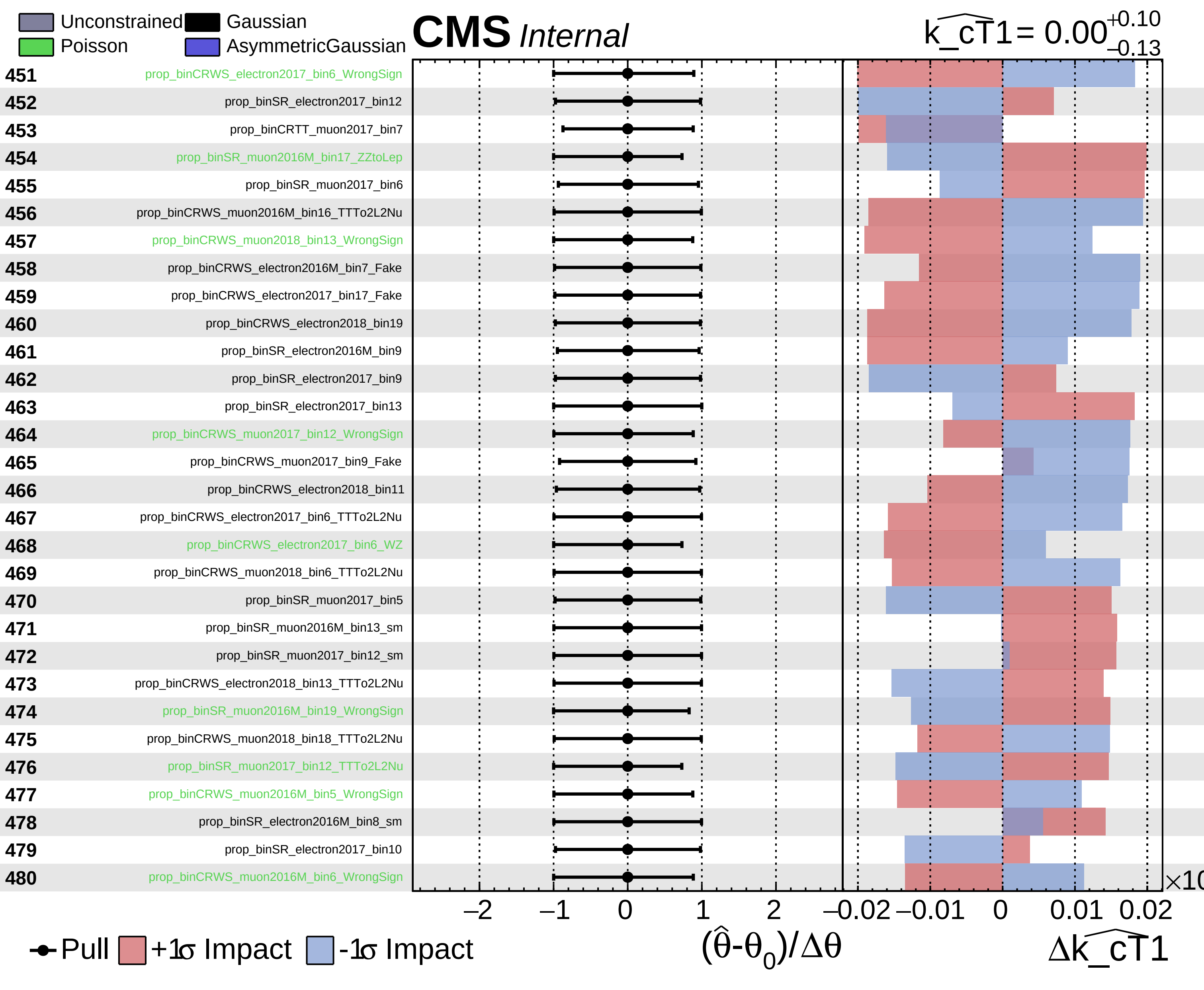


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
361 prop_binSR_electron2016M_bin17_WrongSign
362 prop_binCRWS_muon2017_bin6_Fake
363 prop_binCRTT_electron2016M_bin10
364 prop_binSR_muon2017_bin3
365 prop_binSR_muon2017_bin11
366 prop_binSR_muon2017_bin18
367 prop_binCRWS_electron2018_bin8
368 prop_binCRWS_muon2017_bin16
369 prop_binSR_electron2017_bin17_sm
370 prop_binCRWS_muon2018_bin12_WrongSign
371 prop_binSR_electron2016M_bin17_WZ
372 prop_binSR_electron2016M_bin12
373 prop_binSR_muon2017_bin13
374 prop_binSR_electron2017_bin18_sm
375 prop_binCRTT_muon2017_bin15
376 prop_binCRWS_muon2018_bin6_Fake
377 prop_binCRWS_electron2018_bin6
378 prop_binSR_muon2016M_bin6
379 prop_binSR_electron2018_bin17
380 prop_binCRTT_electron2016M_bin8
381 prop_binSR_electron2018_bin5
382 prop_binCRWS_electron2017_bin12
383 prop_binSR_electron2017_bin14
384 prop_binCRWS_muon2017_bin20
385 prop_binCRWS_electron2018_bin9
386 prop_binCRWS_muon2016M_bin3_WrongSign
387 prop_binCRWS_electron2018_bin14_Fake
388 prop_binCRTT_muon2017_bin10
389 prop_binSR_electron2016M_bin3
390 prop_binSR_electron2016M_bin11
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

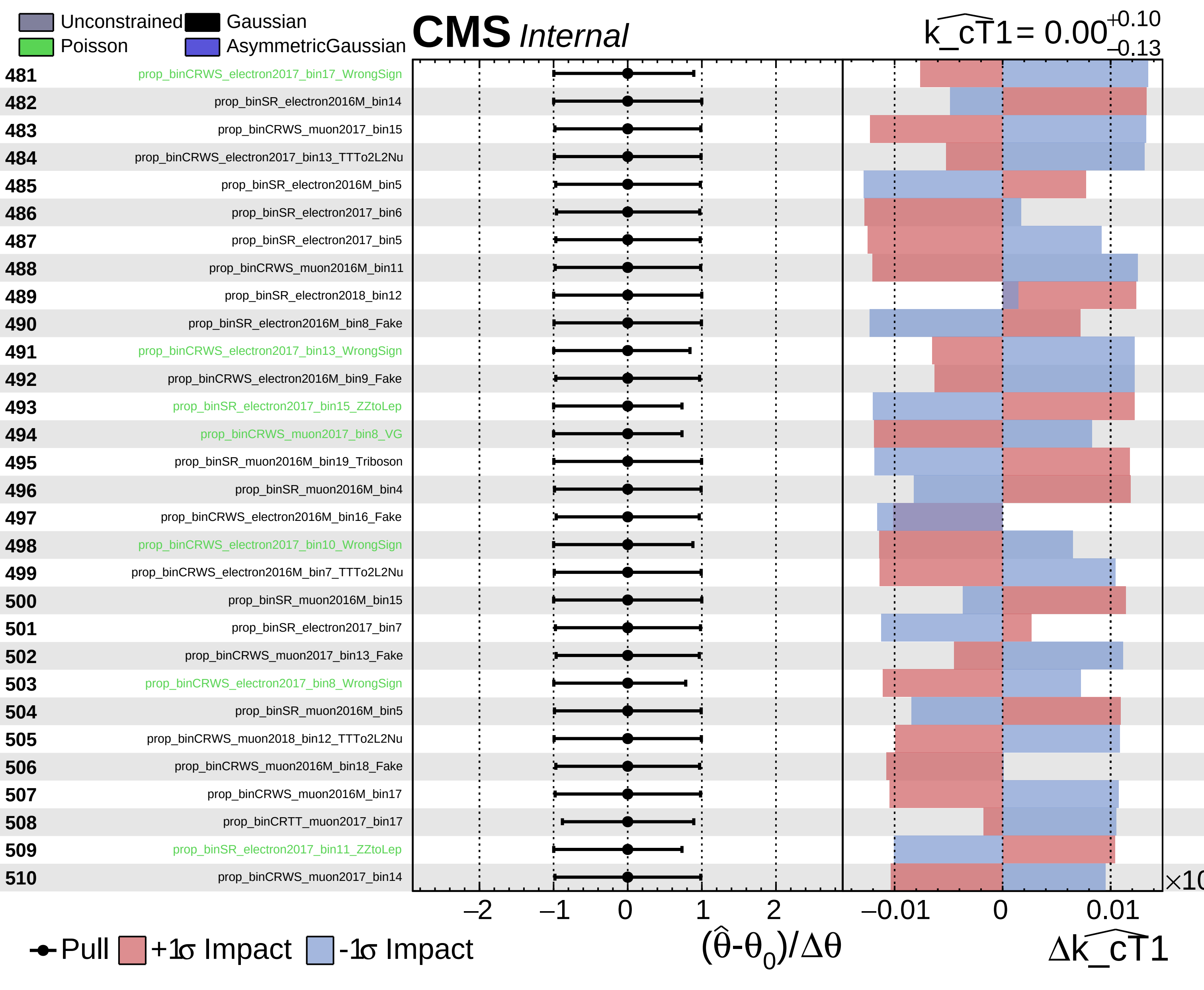

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
391 prop_binSR_electron2018_bin16_sm
392 prop_binCRWS_muon2016M_bin15
393 prop_binCRTT_electron2016M_bin12
394 prop_binSR_electron2018_bin16_TTTo2L2Nu
395 prop_binSR_muon2017_bin16_sm
396 prop_binCRTT_electron2018_bin13
397 prop_binCRWS_muon2017_bin18
398 prop_binCRWS_electron2017_bin19
399 prop_binSR_muon2016M_bin12
400 prop_binSR_electron2016M_bin10
401 prop_binCRWS_electron2017_bin20
402 prop_binSR_muon2016M_bin9
403 prop_binCRWS_muon2016M_bin4_Fake
404 prop_binSR_electron2018_bin3
405 prop_binSR_muon2016M_bin17_sm
406 prop_binCRWS_electron2018_bin7_TTTo2L2Nu
407 prop_binCRWS_muon2018_bin18_Fake
408 prop_binCRWS_electron2017_bin6_Fake
409 prop_binCRWS_muon2016M_bin3_TTTo2L2Nu
410 prop_binCRWS_muon2016M_bin16_WrongSign
411 prop_binCRWS_muon2016M_bin14
412 prop_binCRWS_muon2018_bin10_Fake
413 prop_binSR_electron2016M_bin4
414 prop_binCRWS_electron2016M_bin20_TTTo2L2Nu
415 prop_binCRWS_electron2018_bin15
416 prop_binCRTT_electron2016M_bin11
417 prop_binCRTT_electron2017_bin12
418 prop_binSR_muon2016M_bin8
419 prop_binSR_electron2016M_bin16_sm
420 prop_binSR_electron2016M_bin17_sm
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



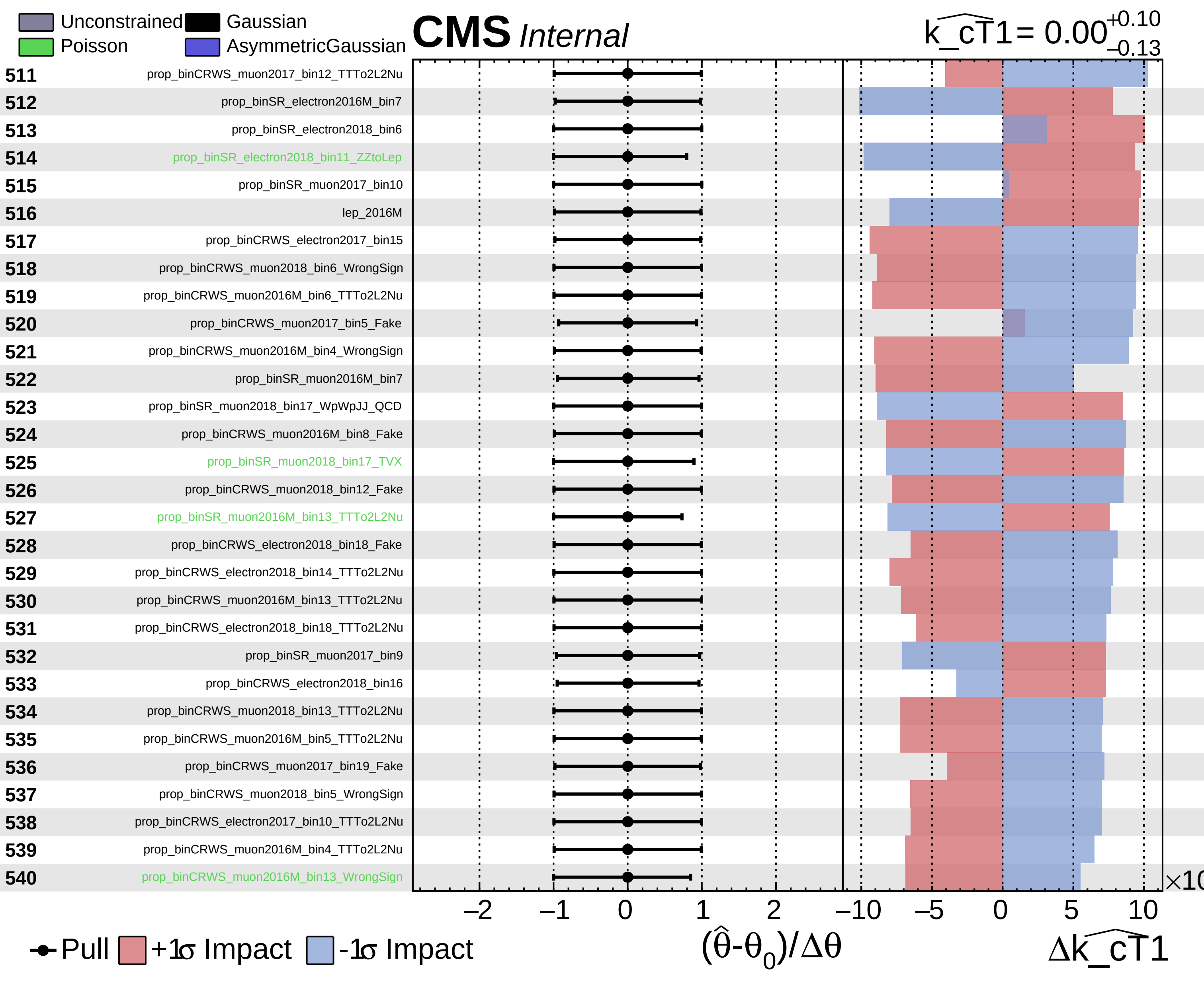
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
421 prop_binSR_electron2016M_bin2
422 prop_binCRTT_electron2016M_bin18
423 prop_binCRWS_electron2018_bin18_VG
424 prop_binCRWS_muon2017_bin7_Fake
425 prop_binCRTT_electron2016M_bin14
426 prop_binSR_electron2016M_bin17_TTTo2L2Nu
427 prop_binCRTT_electron2016M_bin4
428 prop_binCRWS_electron2016M_bin18
429 prop_binCRWS_electron2018_bin7_WrongSign
430 prop_binCRTT_electron2018_bin15
431 prop_binCRWS_electron2016M_bin8_Fake
432 prop_binSR_muon2017_bin8
433 prop_binSR_electron2017_bin17_ZZtoLep
434 prop_binCRWS_electron2016M_bin12
435 prop_binSR_electron2017_bin18_WrongSign
436 prop_binCRWS_muon2017_bin11
437 prop_binCRWS_muon2016M_bin12
438 prop_binSR_electron2017_bin11_sm
439 prop_binCRWS_muon2016M_bin6_Fake
440 prop_binCRWS_electron2018_bin13_Fake
441 prop_binSR_electron2018_bin11_sm
442 prop_binCRWS_muon2017_bin10_Fake
443 prop_binCRWS_electron2018_bin13_WrongSign
444 prop_binCRWS_muon2018_bin18_WrongSign
445 prop_binSR_electron2018_bin14_sm
446 prop_binSR_electron2017_bin15_sm
447 prop_binCRWS_electron2017_bin8_Fake
448 prop_binCRTT_muon2017_bin13
449 prop_binSR_electron2018_bin14_TTTo2L2Nu
450 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



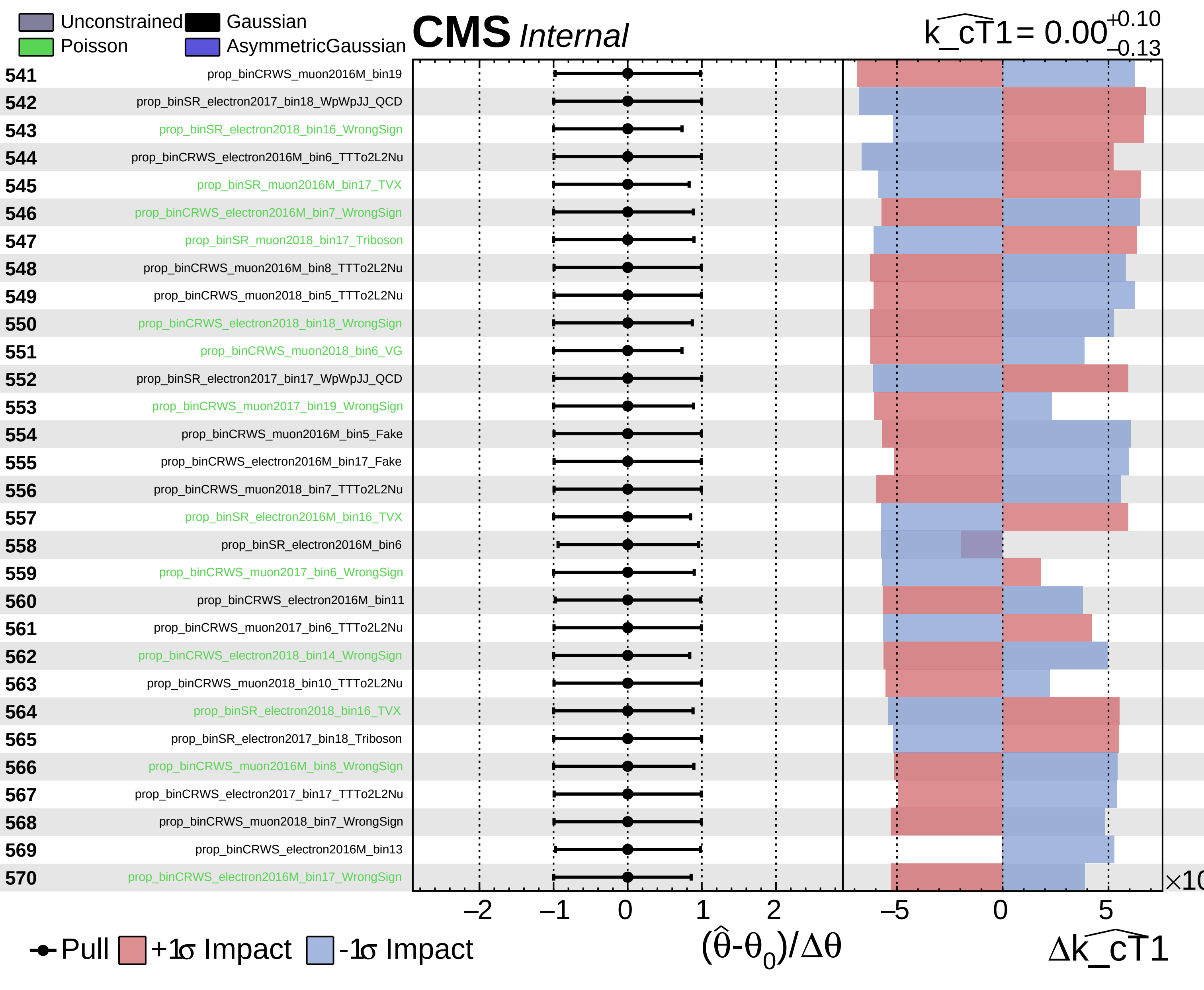
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
451 prop_binCRWS_electron2017_bin6_WrongSign
452 prop_binSR_electron2017_bin12
453 prop_binCRTT_muon2017_bin7
454 prop_binSR_muon2016M_bin17_ZZtoLep
455 prop_binSR_muon2017_bin6
456 prop_binCRWS_muon2016M_bin16_TTTo2L2Nu
457 prop_binCRWS_muon2018_bin13_WrongSign
458 prop_binCRWS_electron2016M_bin7_Fake
459 prop_binCRWS_electron2017_bin17_Fake
460 prop_binCRWS_electron2018_bin19
461 prop_binSR_electron2016M_bin9
462 prop_binSR_electron2017_bin9
463 prop_binSR_electron2017_bin13
464 prop_binCRWS_muon2017_bin12_WrongSign
465 prop_binCRWS_muon2017_bin9_Fake
466 prop_binCRWS_electron2018_bin11
467 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
468 prop_binCRWS_electron2017_bin6_WZ
469 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
470 prop_binSR_muon2017_bin5
471 prop_binSR_muon2016M_bin13_sm
472 prop_binSR_muon2017_bin12_sm
473 prop_binCRWS_electron2018_bin13_TTTo2L2Nu
474 prop_binSR_muon2016M_bin19_WrongSign
475 prop_binCRWS_muon2018_bin18_TTTo2L2Nu
476 prop_binSR_muon2017_bin12_TTTo2L2Nu
477 prop_binCRWS_muon2016M_bin5_WrongSign
478 prop_binSR_electron2016M_bin8_sm
479 prop_binSR_electron2017_bin10
480 prop_binCRWS_muon2016M_bin6_WrongSign
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



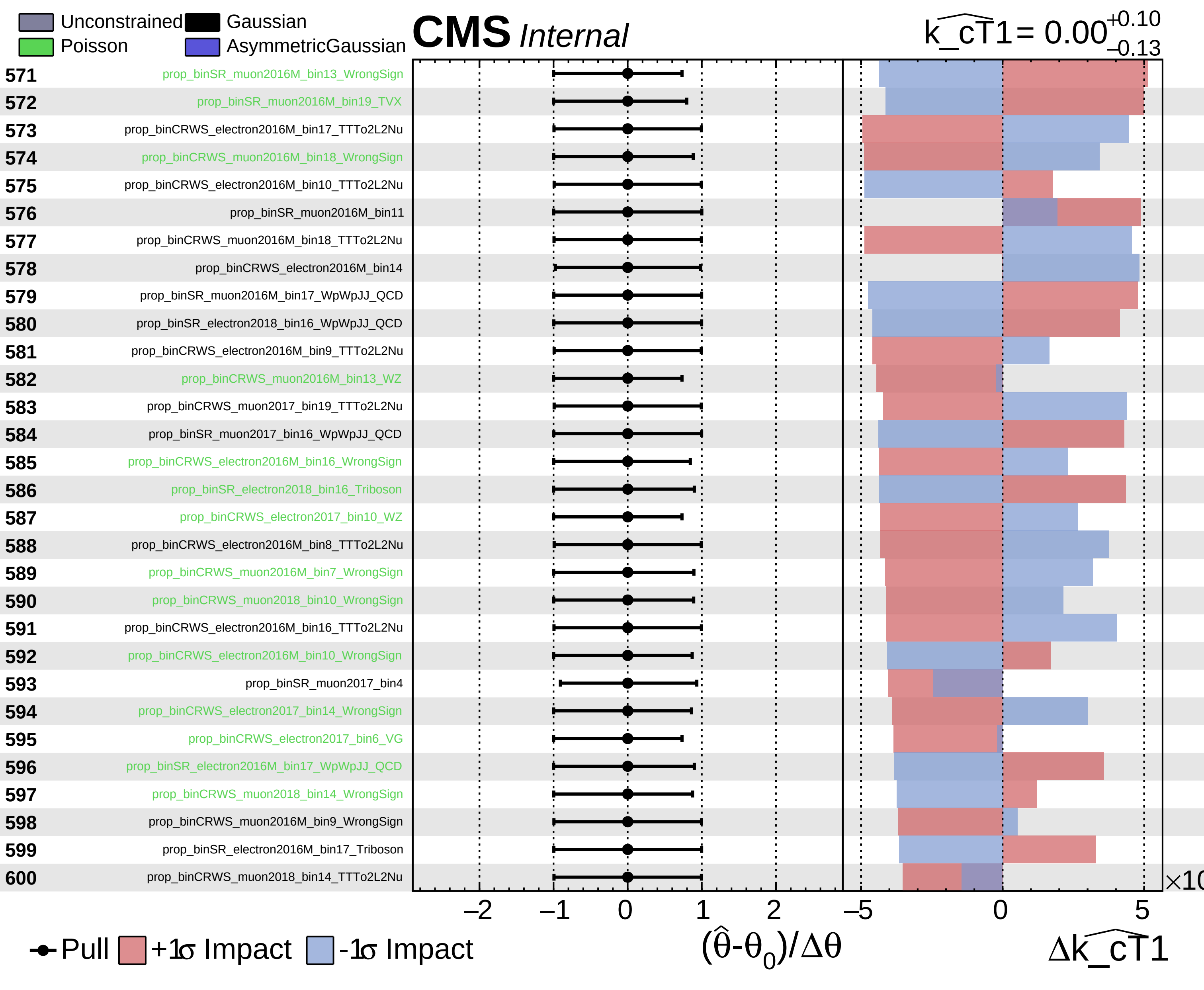
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
481 prop_binCRWS_electron2017_bin17_WrongSign
482 prop_binSR_electron2016M_bin14
483 prop_binCRWS_muon2017_bin15
484 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
485 prop_binSR_electron2016M_bin5
486 prop_binSR_electron2017_bin6
487 prop_binSR_electron2017_bin5
488 prop_binCRWS_muon2016M_bin11
489 prop_binSR_electron2018_bin12
490 prop_binSR_electron2016M_bin8_Fake
491 prop_binCRWS_electron2017_bin13_WrongSign
492 prop_binCRWS_electron2016M_bin9_Fake
493 prop_binSR_electron2017_bin15_ZZtoLep
494 prop_binCRWS_muon2017_bin8_VG
495 prop_binSR_muon2016M_bin19_Triboson
496 prop_binSR_muon2016M_bin4
497 prop_binCRWS_electron2016M_bin16_Fake
498 prop_binCRWS_electron2017_bin10_WrongSign
499 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
500 prop_binSR_muon2016M_bin15
501 prop_binSR_electron2017_bin7
502 prop_binCRWS_muon2017_bin13_Fake
503 prop_binCRWS_electron2017_bin8_WrongSign
504 prop_binSR_muon2016M_bin5
505 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
506 prop_binCRWS_muon2016M_bin18_Fake
507 prop_binCRWS_muon2016M_bin17
508 prop_binCRTT_muon2017_bin17
509 prop_binSR_electron2017_bin11_ZZtoLep
510 prop_binCRWS_muon2017_bin14
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
511 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
512 prop_binSR_electron2016M_bin7
513 prop_binSR_electron2018_bin6
514 prop_binSR_electron2018_bin11_ZZtoLep
515 prop_binSR_muon2017_bin10
516 lep_2016M
517 prop_binCRWS_electron2017_bin15
518 prop_binCRWS_muon2018_bin6_WrongSign
519 prop_binCRWS_muon2016M_bin6_TTTo2L2Nu
520 prop_binCRWS_muon2017_bin5_Fake
521 prop_binCRWS_muon2016M_bin4_WrongSign
522 prop_binSR_muon2016M_bin7
523 prop_binSR_muon2018_bin17_WpWpJJ_QCD
524 prop_binCRWS_muon2016M_bin8_Fake
525 prop_binSR_muon2018_bin17_TVX
526 prop_binCRWS_muon2018_bin12_Fake
527 prop_binSR_muon2016M_bin13_TTTo2L2Nu
528 prop_binCRWS_electron2018_bin18_Fake
529 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
530 prop_binCRWS_muon2016M_bin13_TTTo2L2Nu
531 prop_binCRWS_electron2018_bin18_TTTo2L2Nu
532 prop_binSR_muon2017_bin9
533 prop_binCRWS_electron2018_bin16
534 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
535 prop_binCRWS_muon2016M_bin5_TTTo2L2Nu
536 prop_binCRWS_muon2017_bin19_Fake
537 prop_binCRWS_muon2018_bin5_WrongSign
538 prop_binCRWS_electron2017_bin10_TTTo2L2Nu
539 prop_binCRWS_muon2016M_bin4_TTTo2L2Nu
540 prop_binCRWS_muon2016M_bin13_WrongSign
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
541 prop_binCRWS_muon2016M_bin19
542 prop_binSR_electron2017_bin18_WpWpJJ_QCD
543 prop_binSR_electron2018_bin16_WrongSign
544 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
545 prop_binSR_muon2016M_bin17_TVX
546 prop_binCRWS_electron2016M_bin7_WrongSign
547 prop_binSR_muon2018_bin17_Triboson
548 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
549 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
550 prop_binCRWS_electron2018_bin18_WrongSign
551 prop_binCRWS_muon2018_bin6_VG
552 prop_binSR_electron2017_bin17_WpWpJJ_QCD
553 prop_binCRWS_muon2017_bin19_WrongSign
554 prop_binCRWS_muon2016M_bin5_Fake
555 prop_binCRWS_electron2016M_bin17_Fake
556 prop_binCRWS_muon2018_bin7_TTTo2L2Nu
557 prop_binSR_electron2016M_bin16_TVX
558 prop_binSR_electron2016M_bin6
559 prop_binCRWS_muon2017_bin6_WrongSign
560 prop_binCRWS_electron2016M_bin11
561 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
562 prop_binCRWS_electron2018_bin14_WrongSign
563 prop_binCRWS_muon2018_bin10_TTTo2L2Nu
564 prop_binSR_electron2018_bin16_TVX
565 prop_binSR_electron2017_bin18_Triboson
566 prop_binCRWS_muon2016M_bin8_WrongSign
567 prop_binCRWS_electron2017_bin17_TTTo2L2Nu
568 prop_binCRWS_muon2018_bin7_WrongSign
569 prop_binCRWS_electron2016M_bin13
570 prop_binCRWS_electron2016M_bin17_WrongSign
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
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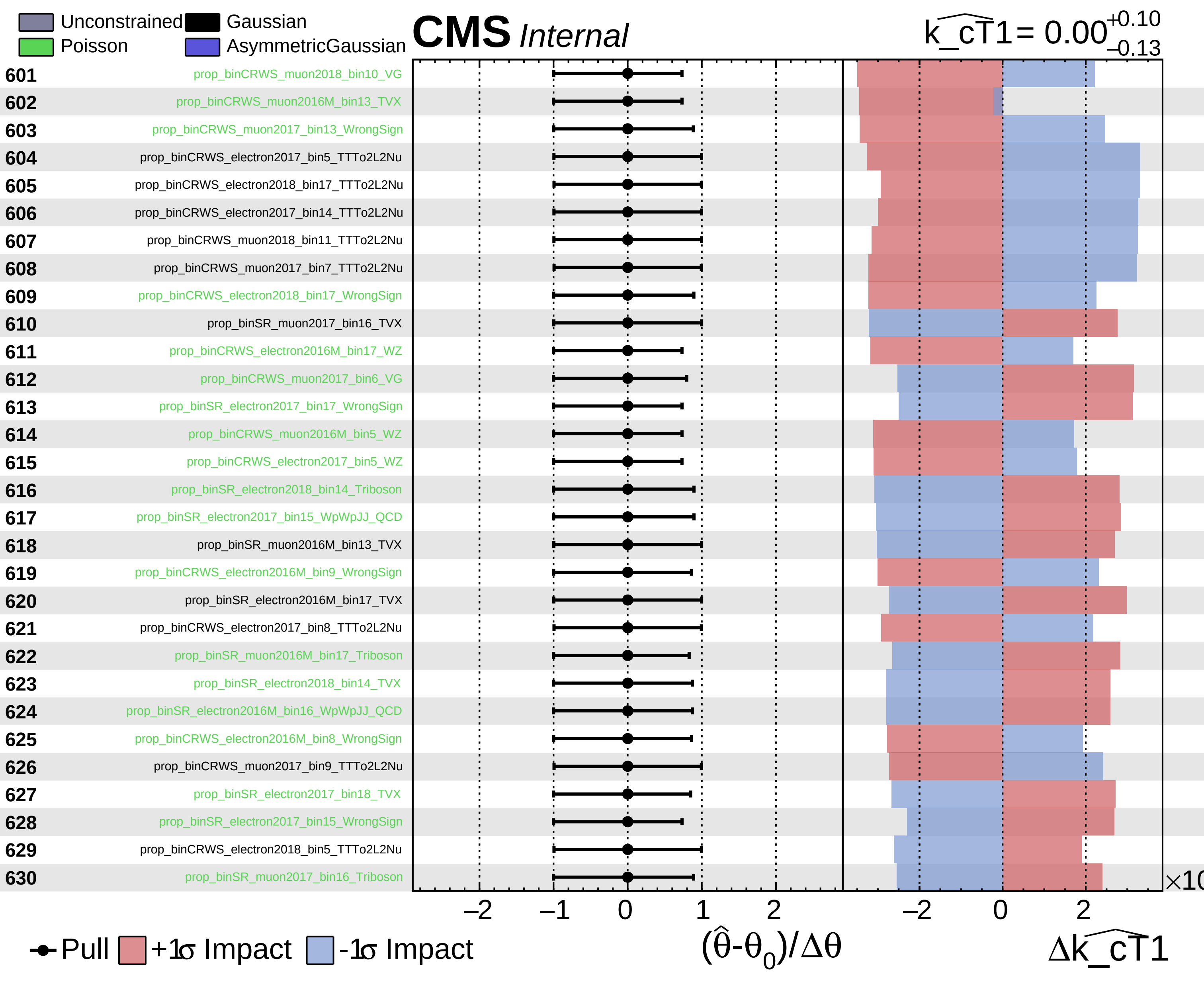
572 prop_binSR_muon2016M_bin19_TVX
573 prop_binCRWS_electron2016M_bin17_TTTo2L2Nu
574 prop_binCRWS_muon2016M_bin18_WrongSign
575 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
576 prop_binSR_muon2016M_bin11
577 prop_binCRWS_muon2016M_bin18_TTTo2L2Nu
578 prop_binCRWS_electron2016M_bin14
579 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
580 prop_binSR_electron2018_bin16_WpWpJJ_QCD
581 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
582 prop_binCRWS_muon2016M_bin13_WZ
583 prop_binCRWS_muon2017_bin19_TTTo2L2Nu
584 prop_binSR_muon2017_bin16_WpWpJJ_QCD
585 prop_binCRWS_electron2016M_bin16_WrongSign
586 prop_binSR_electron2018_bin16_Triboson
587 prop_binCRWS_electron2017_bin10_WZ
588 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
589 prop_binCRWS_muon2016M_bin7_WrongSign
590 prop_binCRWS_muon2018_bin10_WrongSign
591 prop_binCRWS_electron2016M_bin16_TTTo2L2Nu
592 prop_binCRWS_electron2016M_bin10_WrongSign
593 prop_binSR_muon2017_bin4
594 prop_binCRWS_electron2017_bin14_WrongSign
595 prop_binCRWS_electron2017_bin6_VG
596 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
597 prop_binCRWS_muon2018_bin14_WrongSign
598 prop_binCRWS_muon2016M_bin9_WrongSign
599 prop_binSR_electron2016M_bin17_Triboson
600 prop_binCRWS_muon2018_bin14_TTTo2L2Nu
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
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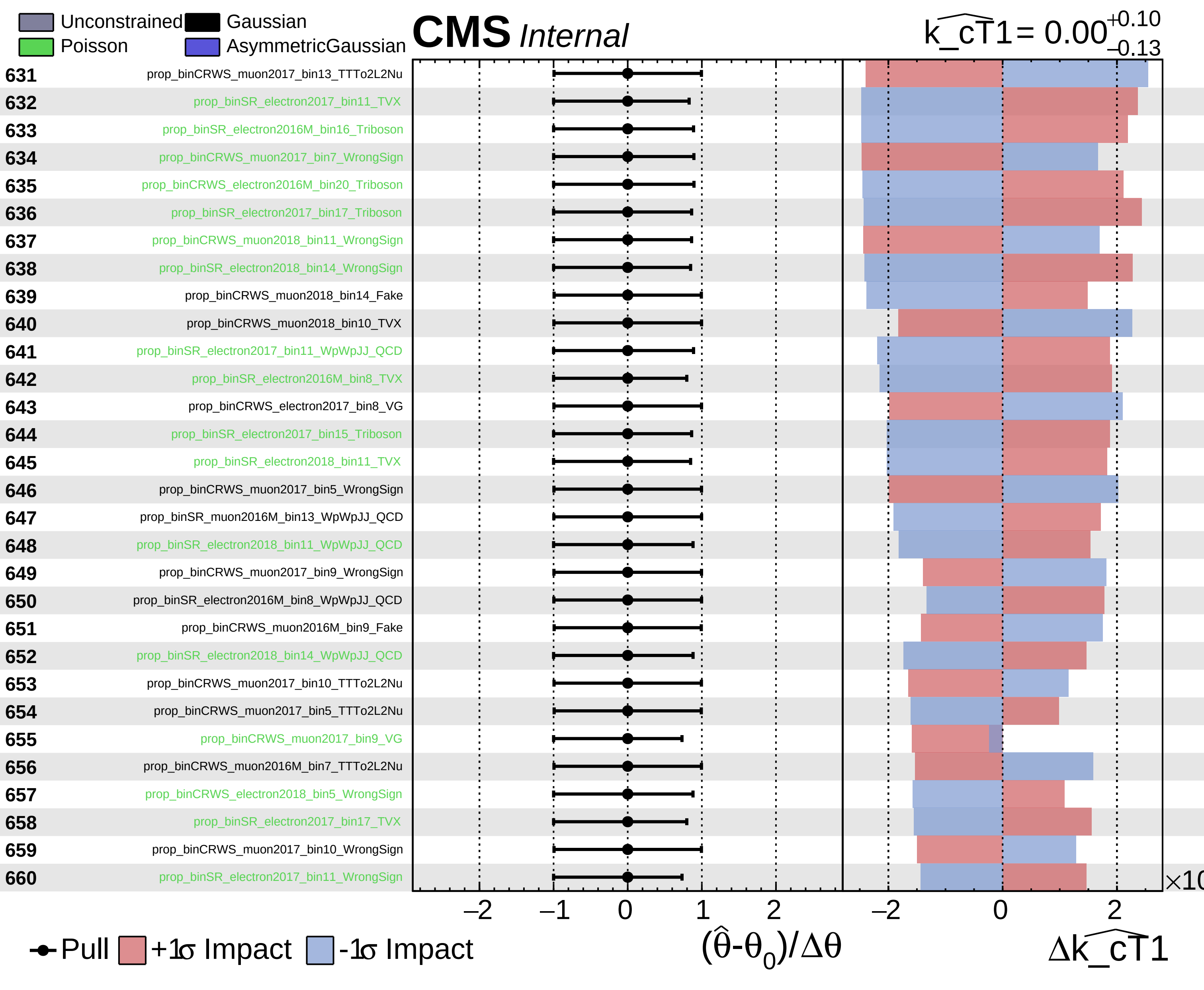
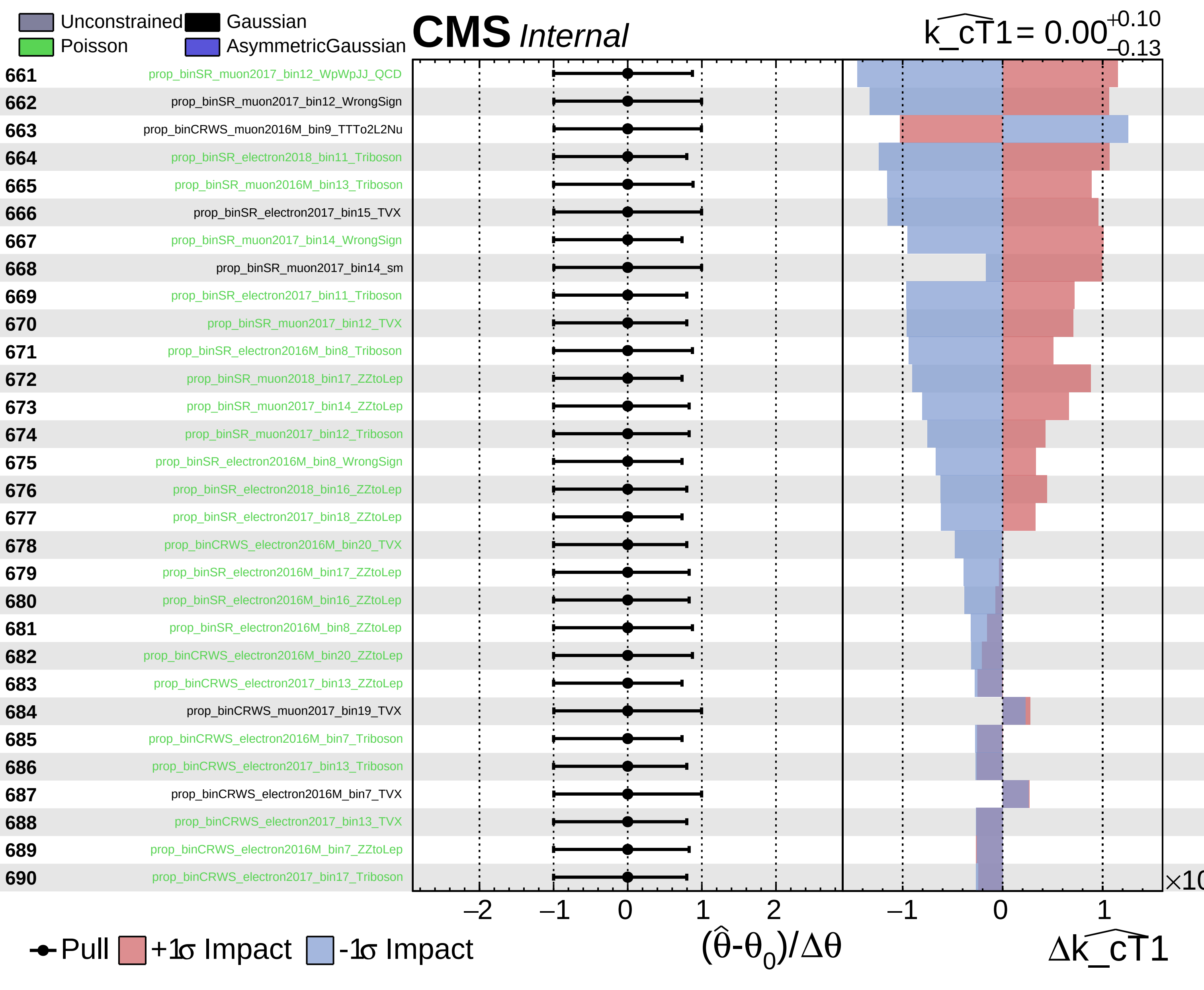
602 prop_binCRWS_muon2016M_bin13_TVX
603 prop_binCRWS_muon2017_bin13_WrongSign
604 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
605 prop_binCRWS_electron2018_bin17_TTTo2L2Nu
606 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
607 prop_binCRWS_muon2018_bin11_TTTo2L2Nu
608 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
609 prop_binCRWS_electron2018_bin17_WrongSign
610 prop_binSR_muon2017_bin16_TVX
611 prop_binCRWS_electron2016M_bin17_WZ
612 prop_binCRWS_muon2017_bin6_VG
613 prop_binSR_electron2017_bin17_WrongSign
614 prop_binCRWS_muon2016M_bin5_WZ
615 prop_binCRWS_electron2017_bin5_WZ
616 prop_binSR_electron2018_bin14_Triboson
617 prop_binSR_electron2017_bin15_WpWpJJ_QCD
618 prop_binSR_muon2016M_bin13_TVX
619 prop_binCRWS_electron2016M_bin9_WrongSign
620 prop_binSR_electron2016M_bin17_TVX
621 prop_binCRWS_electron2017_bin8_TTTo2L2Nu
622 prop_binSR_muon2016M_bin17_Triboson
623 prop_binSR_electron2018_bin14_TVX
624 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
625 prop_binCRWS_electron2016M_bin8_WrongSign
626 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
627 prop_binSR_electron2017_bin18_TVX
628 prop_binSR_electron2017_bin15_WrongSign
629 prop_binCRWS_electron2018_bin5_TTTo2L2Nu
630 prop_binSR_muon2017_bin16_Triboson
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
631 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
632 prop_binSR_electron2017_bin11_TVX
633 prop_binSR_electron2016M_bin16_Triboson
634 prop_binCRWS_muon2017_bin7_WrongSign
635 prop_binCRWS_electron2016M_bin20_Triboson
636 prop_binSR_electron2017_bin17_Triboson
637 prop_binCRWS_muon2018_bin11_WrongSign
638 prop_binSR_electron2018_bin14_WrongSign
639 prop_binCRWS_muon2018_bin14_Fake
640 prop_binCRWS_muon2018_bin10_TVX
641 prop_binSR_electron2017_bin11_WpWpJJ_QCD
642 prop_binSR_electron2016M_bin8_TVX
643 prop_binCRWS_electron2017_bin8_VG
644 prop_binSR_electron2017_bin15_Triboson
645 prop_binSR_electron2018_bin11_TVX
646 prop_binCRWS_muon2017_bin5_WrongSign
647 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
648 prop_binSR_electron2018_bin11_WpWpJJ_QCD
649 prop_binCRWS_muon2017_bin9_WrongSign
650 prop_binSR_electron2016M_bin8_WpWpJJ_QCD
651 prop_binCRWS_muon2016M_bin9_Fake
652 prop_binSR_electron2018_bin14_WpWpJJ_QCD
653 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
654 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
655 prop_binCRWS_muon2017_bin9_VG
656 prop_binCRWS_muon2016M_bin7_TTTo2L2Nu
657 prop_binCRWS_electron2018_bin5_WrongSign
658 prop_binSR_electron2017_bin17_TVX
659 prop_binCRWS_muon2017_bin10_WrongSign
660 prop_binSR_electron2017_bin11_WrongSign
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
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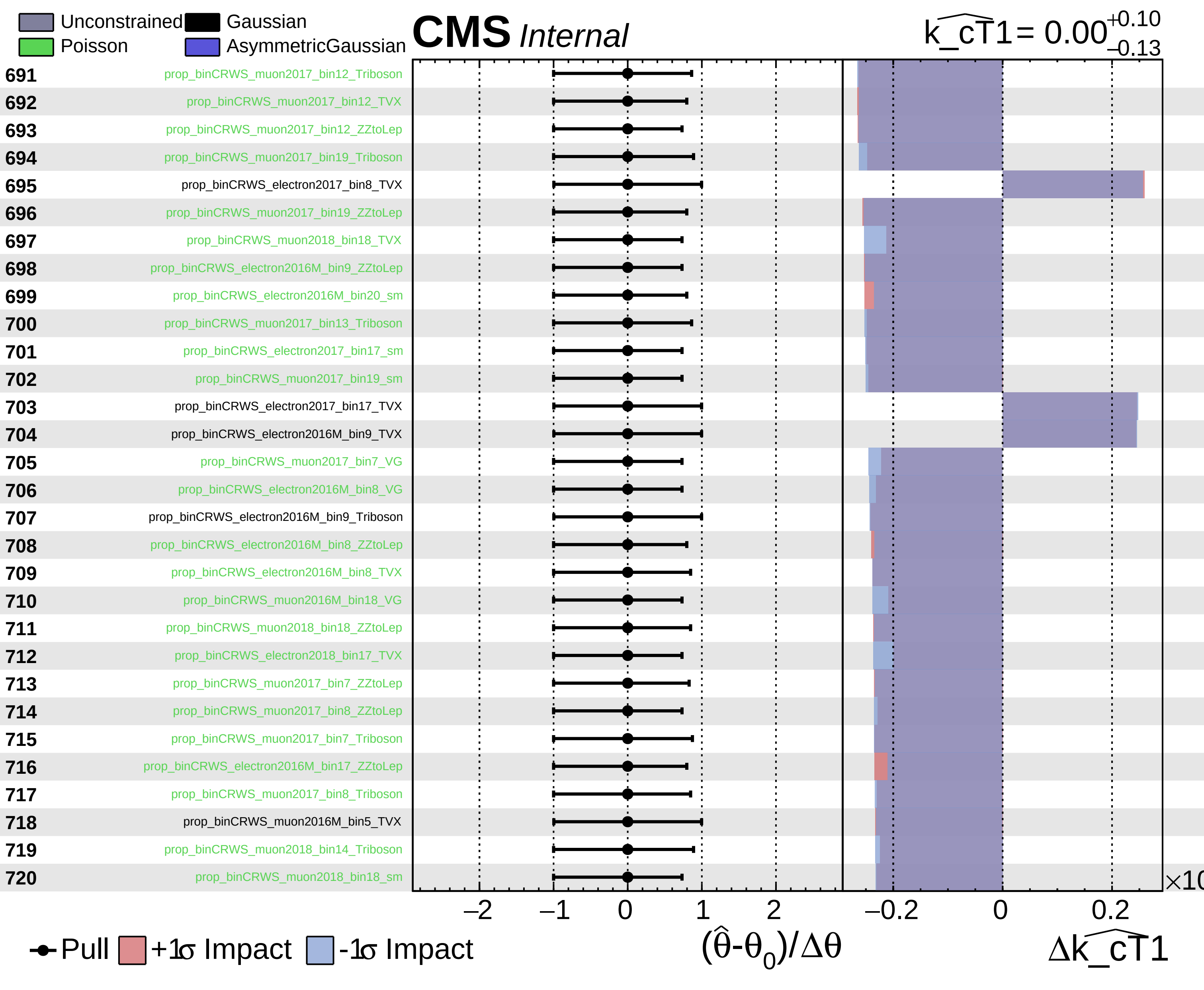
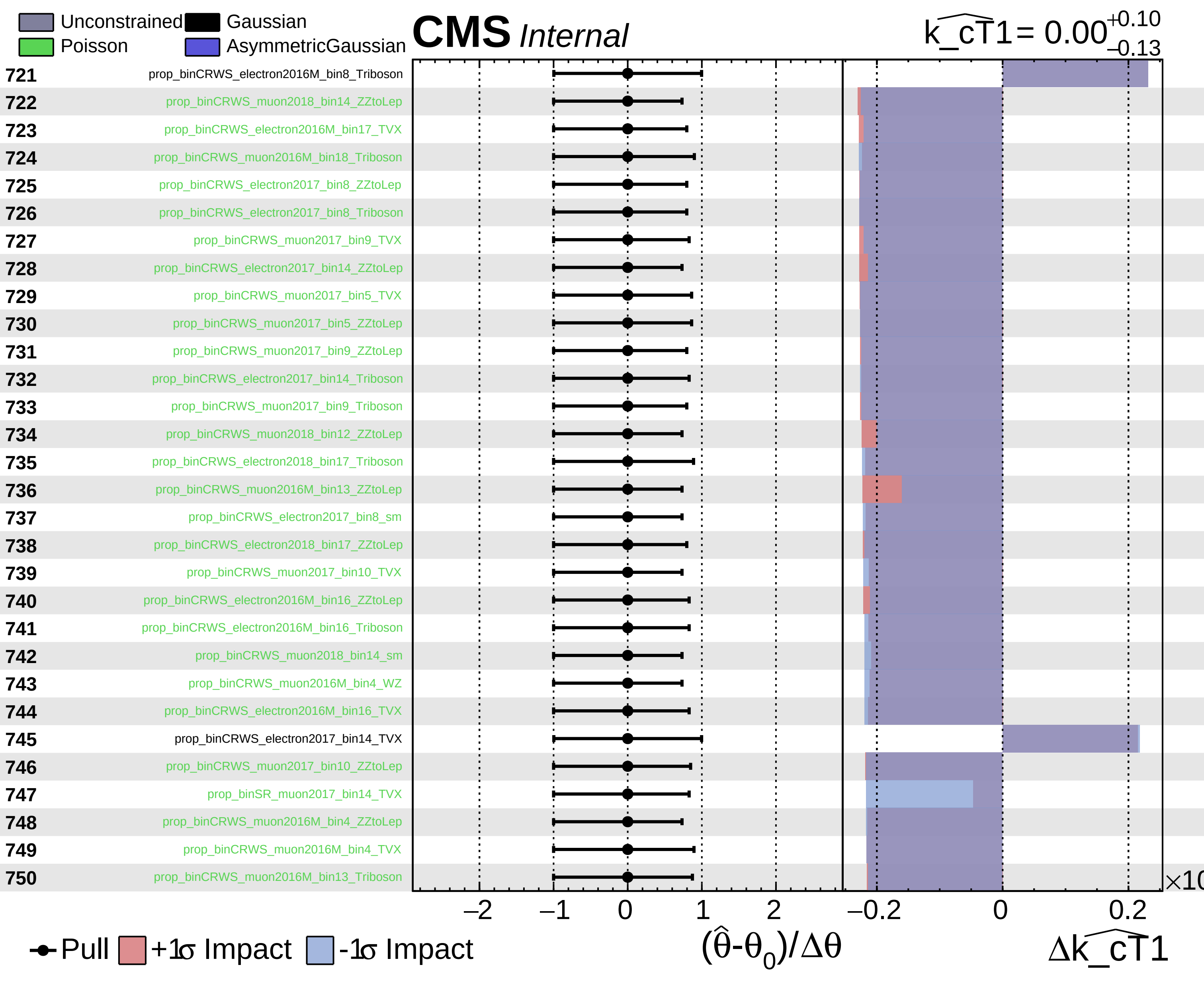
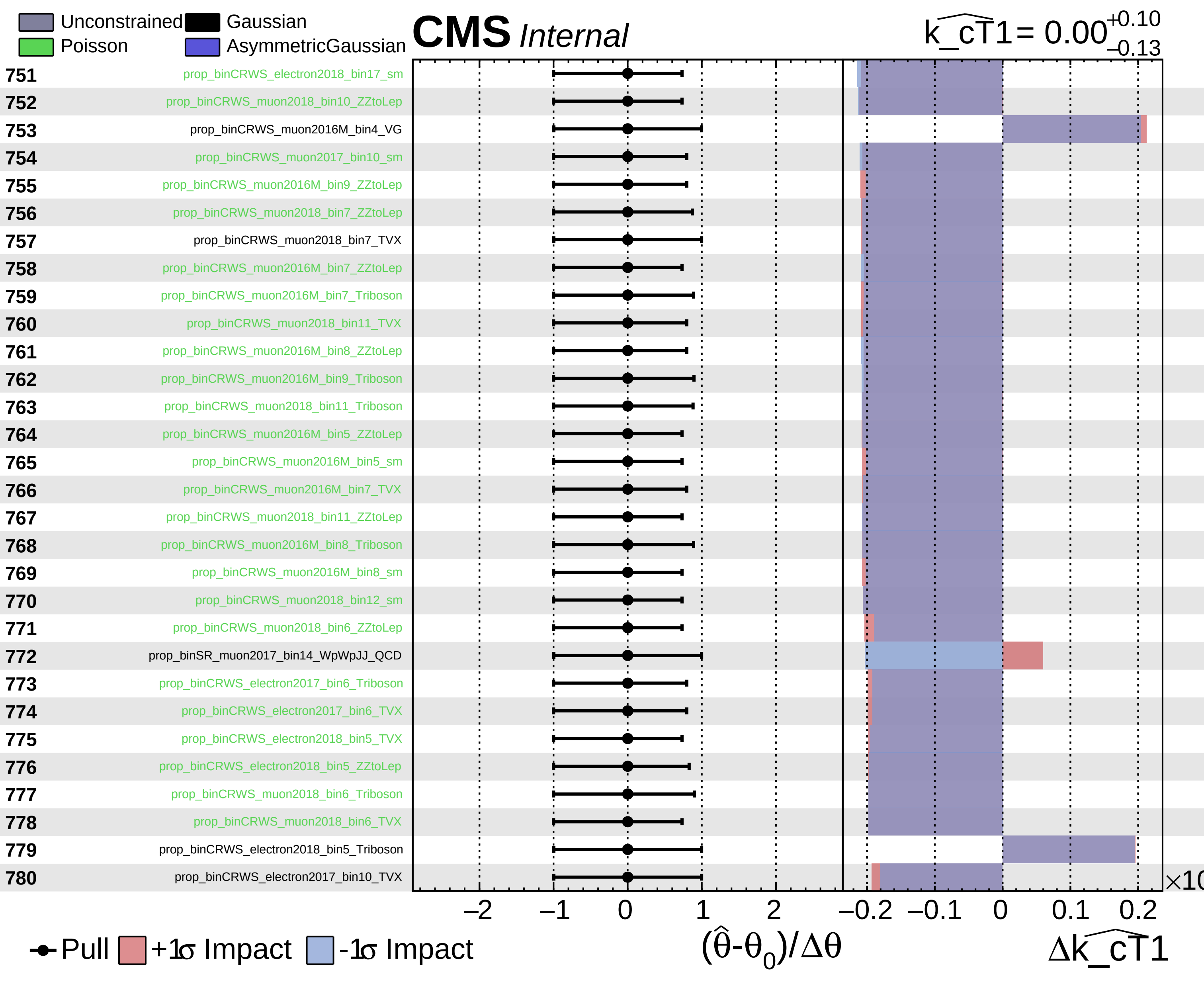
662 prop_binSR_muon2017_bin12_WrongSign
663 prop_binCRWS_muon2016M_bin9_TTTo2L2Nu
664 prop_binSR_electron2018_bin11_Triboson
665 prop_binSR_muon2016M_bin13_Triboson
666 prop_binSR_electron2017_bin15_TVX
667 prop_binSR_muon2017_bin14_WrongSign
668 prop_binSR_muon2017_bin14_sm
669 prop_binSR_electron2017_bin11_Triboson
670 prop_binSR_muon2017_bin12_TVX
671 prop_binSR_electron2016M_bin8_Triboson
672 prop_binSR_muon2018_bin17_ZZtoLep
673 prop_binSR_muon2017_bin14_ZZtoLep
674 prop_binSR_muon2017_bin12_Triboson
675 prop_binSR_electron2016M_bin8_WrongSign
676 prop_binSR_electron2018_bin16_ZZtoLep
677 prop_binSR_electron2017_bin18_ZZtoLep
678 prop_binCRWS_electron2016M_bin20_TVX
679 prop_binSR_electron2016M_bin17_ZZtoLep
680 prop_binSR_electron2016M_bin16_ZZtoLep
681 prop_binSR_electron2016M_bin8_ZZtoLep
682 prop_binCRWS_electron2016M_bin20_ZZtoLep
683 prop_binCRWS_electron2017_bin13_ZZtoLep
684 prop_binCRWS_muon2017_bin19_TVX
685 prop_binCRWS_electron2016M_bin7_Triboson
686 prop_binCRWS_electron2017_bin13_Triboson
687 prop_binCRWS_electron2016M_bin7_TVX
688 prop_binCRWS_electron2017_bin13_TVX
689 prop_binCRWS_electron2016M_bin7_ZZtoLep
690 prop_binCRWS_electron2017_bin17_Triboson
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



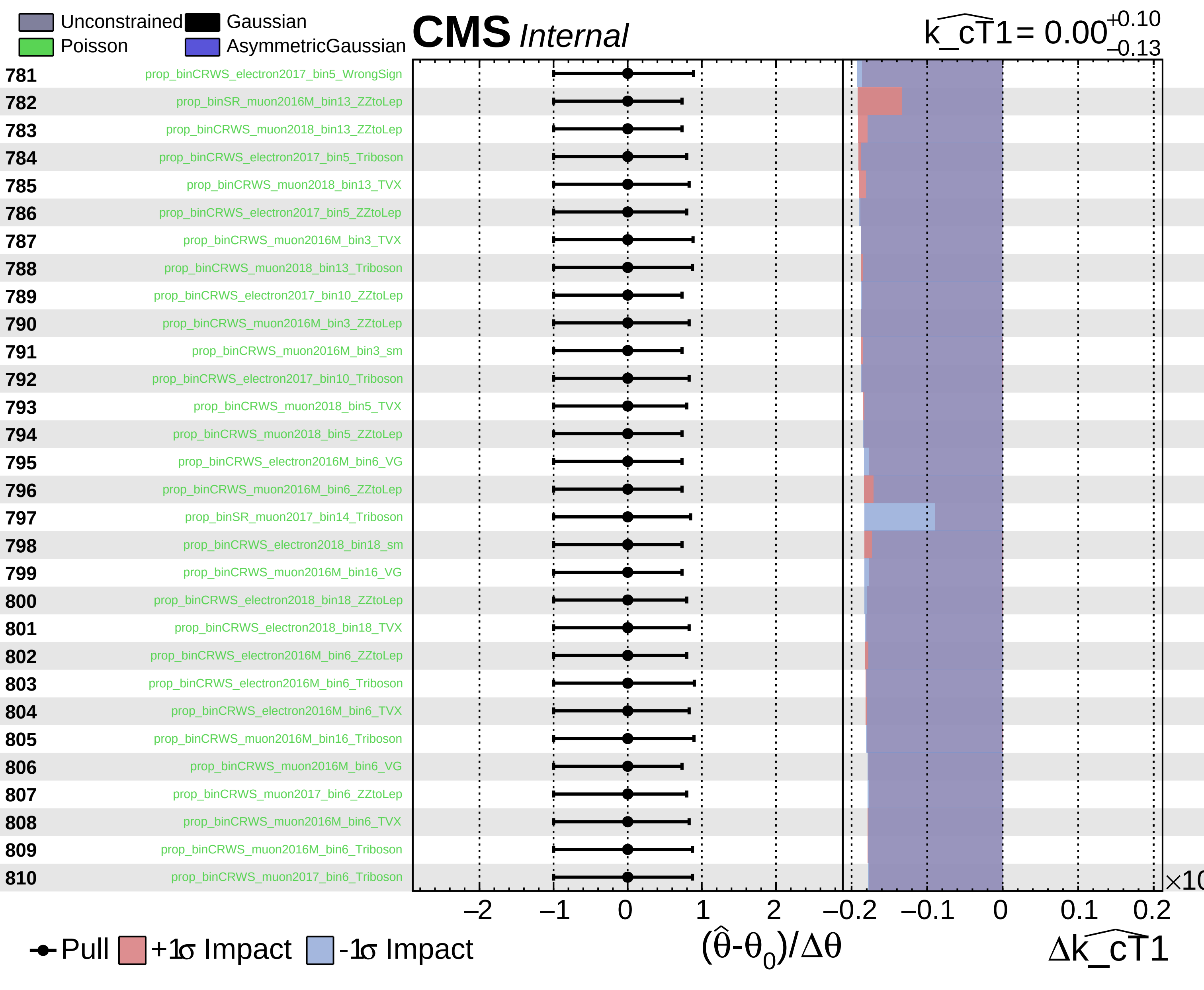
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
691 prop_binCRWS_muon2017_bin12_Triboson
692 prop_binCRWS_muon2017_bin12_TVX
693 prop_binCRWS_muon2017_bin12_ZZtoLep
694 prop_binCRWS_muon2017_bin19_Triboson
695 prop_binCRWS_electron2017_bin8_TVX
696 prop_binCRWS_muon2017_bin19_ZZtoLep
697 prop_binCRWS_muon2018_bin18_TVX
698 prop_binCRWS_electron2016M_bin9_ZZtoLep
699 prop_binCRWS_electron2016M_bin20_sm
700 prop_binCRWS_muon2017_bin13_Triboson
701 prop_binCRWS_electron2017_bin17_sm
702 prop_binCRWS_muon2017_bin19_sm
703 prop_binCRWS_electron2017_bin17_TVX
704 prop_binCRWS_electron2016M_bin9_TVX
705 prop_binCRWS_muon2017_bin7_VG
706 prop_binCRWS_electron2016M_bin8_VG
707 prop_binCRWS_electron2016M_bin9_Triboson
708 prop_binCRWS_electron2016M_bin8_ZZtoLep
709 prop_binCRWS_electron2016M_bin8_TVX
710 prop_binCRWS_muon2016M_bin18_VG
711 prop_binCRWS_muon2018_bin18_ZZtoLep
712 prop_binCRWS_electron2018_bin17_TVX
713 prop_binCRWS_muon2017_bin7_ZZtoLep
714 prop_binCRWS_muon2017_bin8_ZZtoLep
715 prop_binCRWS_muon2017_bin7_Triboson
716 prop_binCRWS_electron2016M_bin17_ZZtoLep
717 prop_binCRWS_muon2017_bin8_Triboson
718 prop_binCRWS_muon2016M_bin5_TVX
719 prop_binCRWS_muon2018_bin14_Triboson
720 prop_binCRWS_muon2018_bin18_sm
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

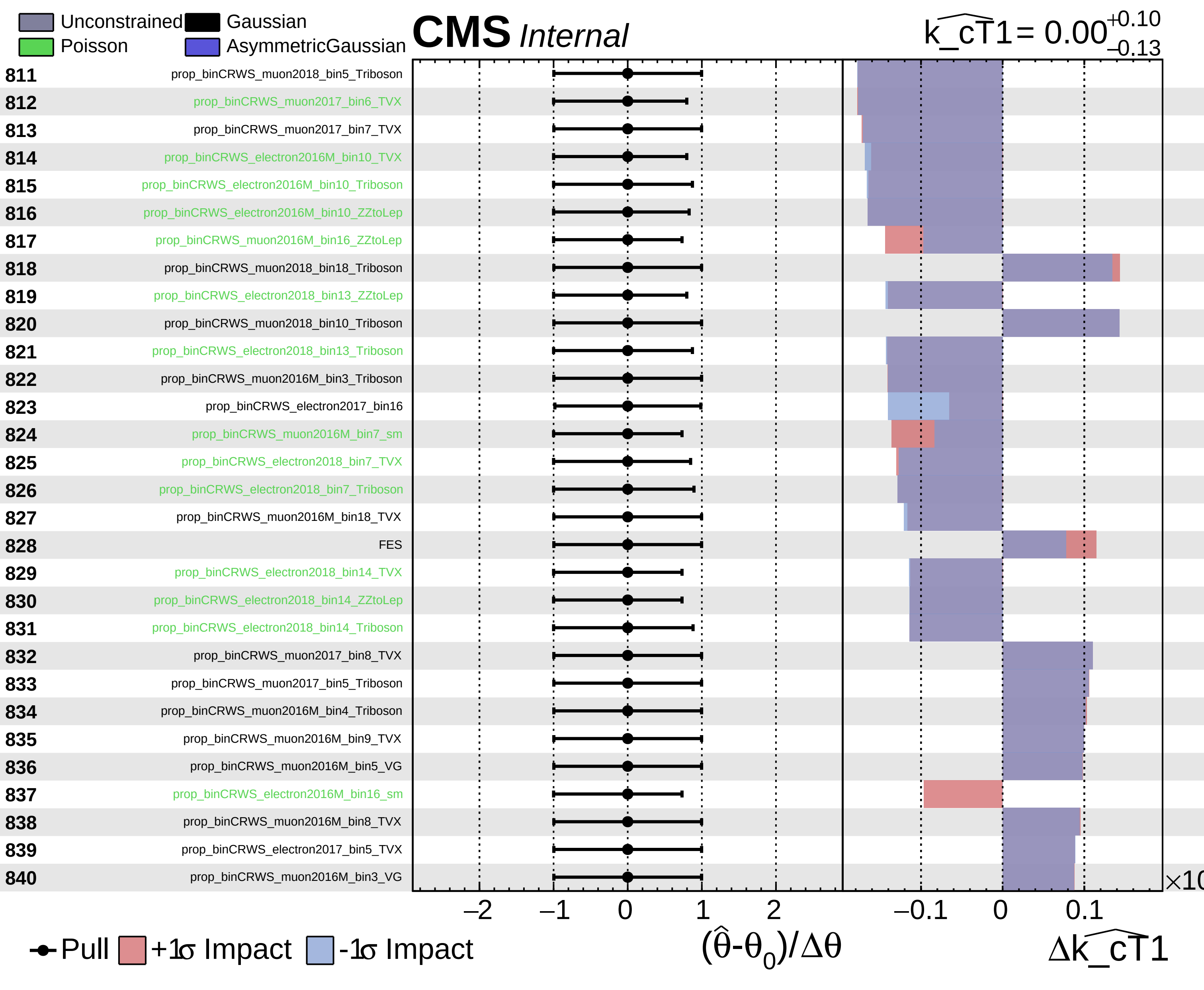


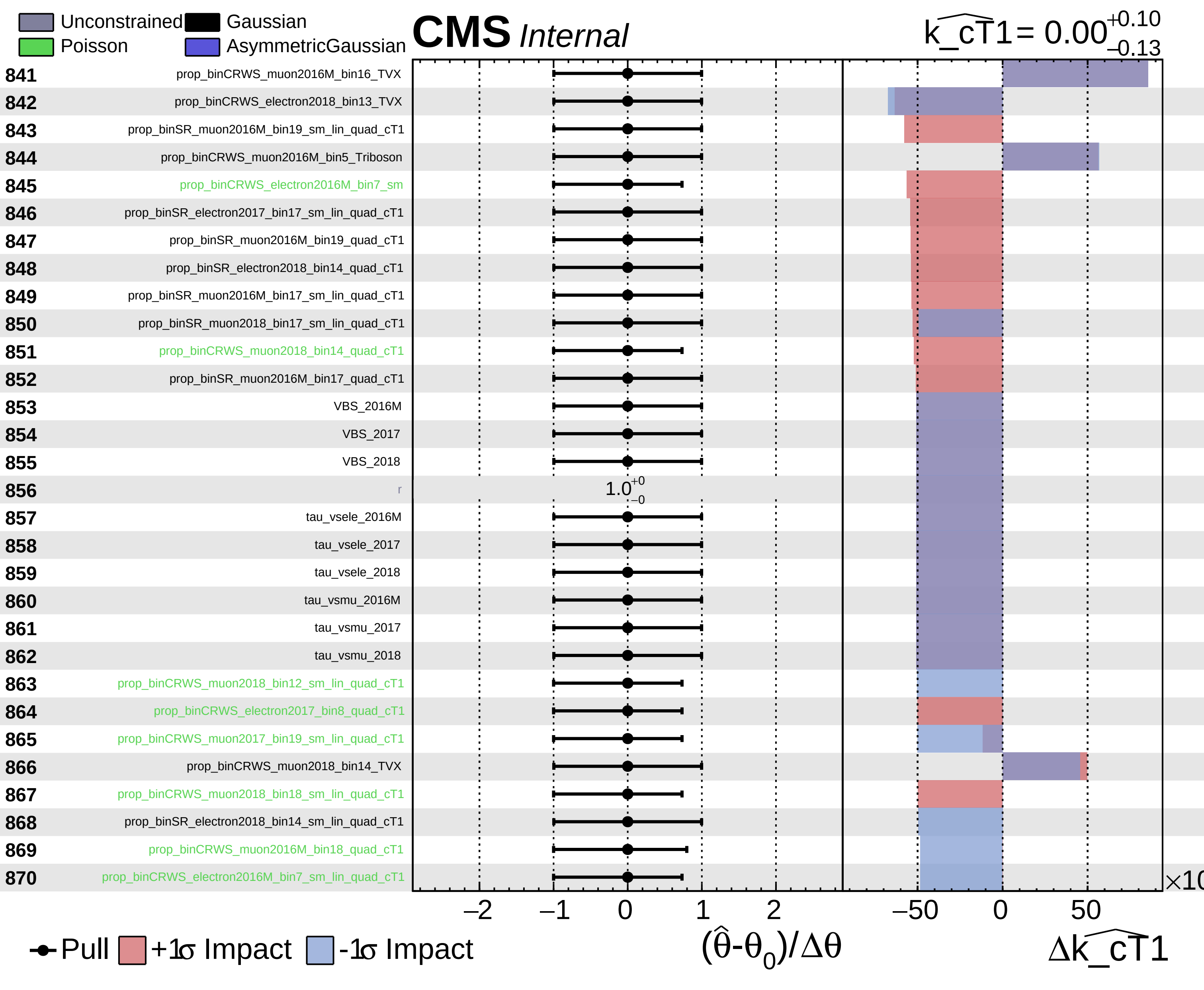
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
721 prop_binCRWS_electron2016M_bin8_Triboson
722 prop_binCRWS_muon2018_bin14_ZZtoLep
723 prop_binCRWS_electron2016M_bin17_TVX
724 prop_binCRWS_muon2016M_bin18_Triboson
725 prop_binCRWS_electron2017_bin8_ZZtoLep
726 prop_binCRWS_electron2017_bin8_Triboson
727 prop_binCRWS_muon2017_bin9_TVX
728 prop_binCRWS_electron2017_bin14_ZZtoLep
729 prop_binCRWS_muon2017_bin5_TVX
730 prop_binCRWS_muon2017_bin5_ZZtoLep
731 prop_binCRWS_muon2017_bin9_ZZtoLep
732 prop_binCRWS_electron2017_bin14_Triboson
733 prop_binCRWS_muon2017_bin9_Triboson
734 prop_binCRWS_muon2018_bin12_ZZtoLep
735 prop_binCRWS_electron2018_bin17_Triboson
736 prop_binCRWS_muon2016M_bin13_ZZtoLep
737 prop_binCRWS_electron2017_bin8_sm
738 prop_binCRWS_electron2018_bin17_ZZtoLep
739 prop_binCRWS_muon2017_bin10_TVX
740 prop_binCRWS_electron2016M_bin16_ZZtoLep
741 prop_binCRWS_electron2016M_bin16_Triboson
742 prop_binCRWS_muon2018_bin14_sm
743 prop_binCRWS_muon2016M_bin4_WZ
744 prop_binCRWS_electron2016M_bin16_TVX
745 prop_binCRWS_electron2017_bin14_TVX
746 prop_binCRWS_muon2017_bin10_ZZtoLep
747 prop_binSR_muon2017_bin14_TVX
748 prop_binCRWS_muon2016M_bin4_ZZtoLep
749 prop_binCRWS_muon2016M_bin4_TVX
750 prop_binCRWS_muon2016M_bin13_Triboson
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1



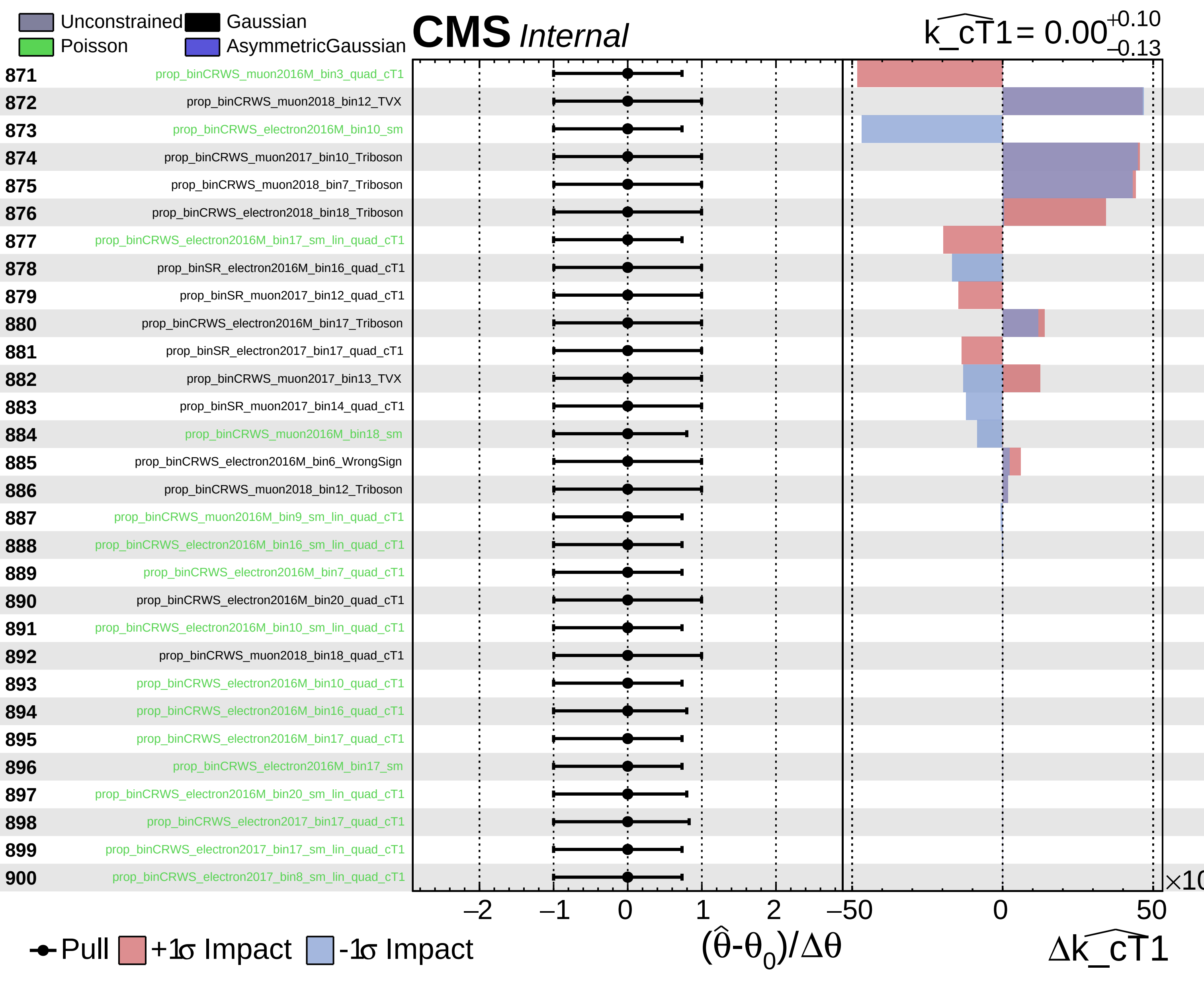
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
751 prop_binCRWS_electron2018_bin17_sm
752 prop_binCRWS_muon2018_bin10_ZZtoLep
753 prop_binCRWS_muon2016M_bin4_VG
754 prop_binCRWS_muon2017_bin10_sm
755 prop_binCRWS_muon2016M_bin9_ZZtoLep
756 prop_binCRWS_muon2018_bin7_ZZtoLep
757 prop_binCRWS_muon2018_bin7_TVX
758 prop_binCRWS_muon2016M_bin7_ZZtoLep
759 prop_binCRWS_muon2016M_bin7_Triboson
760 prop_binCRWS_muon2018_bin11_TVX
761 prop_binCRWS_muon2016M_bin8_ZZtoLep
762 prop_binCRWS_muon2016M_bin9_Triboson
763 prop_binCRWS_muon2018_bin11_Triboson
764 prop_binCRWS_muon2016M_bin5_ZZtoLep
765 prop_binCRWS_muon2016M_bin5_sm
766 prop_binCRWS_muon2016M_bin7_TVX
767 prop_binCRWS_muon2018_bin11_ZZtoLep
768 prop_binCRWS_muon2016M_bin8_Triboson
769 prop_binCRWS_muon2016M_bin8_sm
770 prop_binCRWS_muon2018_bin12_sm
771 prop_binCRWS_muon2018_bin6_ZZtoLep
772 prop_binSR_muon2017_bin14_WpWpJJ_QCD
773 prop_binCRWS_electron2017_bin6_Triboson
774 prop_binCRWS_electron2017_bin6_TVX
775 prop_binCRWS_electron2018_bin5_TVX
776 prop_binCRWS_electron2018_bin5_ZZtoLep
777 prop_binCRWS_muon2018_bin6_Triboson
778 prop_binCRWS_muon2018_bin6_TVX
779 prop_binCRWS_electron2018_bin5_Triboson
780 prop_binCRWS_electron2017_bin10_TVX
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
781 prop_binCRWS_electron2017_bin5_WrongSign
782 prop_binSR_muon2016M_bin13_ZZtoLep
783 prop_binCRWS_muon2018_bin13_ZZtoLep
784 prop_binCRWS_electron2017_bin5_Triboson
785 prop_binCRWS_muon2018_bin13_TVX
786 prop_binCRWS_electron2017_bin5_ZZtoLep
787 prop_binCRWS_muon2016M_bin3_TVX
788 prop_binCRWS_muon2018_bin13_Triboson
789 prop_binCRWS_electron2017_bin10_ZZtoLep
790 prop_binCRWS_muon2016M_bin3_ZZtoLep
791 prop_binCRWS_muon2016M_bin3_sm
792 prop_binCRWS_electron2017_bin10_Triboson
793 prop_binCRWS_muon2018_bin5_TVX
794 prop_binCRWS_muon2018_bin5_ZZtoLep
795 prop_binCRWS_electron2016M_bin6_VG
796 prop_binCRWS_muon2016M_bin6_ZZtoLep
797 prop_binSR_muon2017_bin14_Triboson
798 prop_binCRWS_electron2018_bin18_sm
799 prop_binCRWS_muon2016M_bin16_VG
800 prop_binCRWS_electron2018_bin18_ZZtoLep
801 prop_binCRWS_electron2018_bin18_TVX
802 prop_binCRWS_electron2016M_bin6_ZZtoLep
803 prop_binCRWS_electron2016M_bin6_Triboson
804 prop_binCRWS_electron2016M_bin6_TVX
805 prop_binCRWS_muon2016M_bin16_Triboson
806 prop_binCRWS_muon2016M_bin6_VG
807 prop_binCRWS_muon2017_bin6_ZZtoLep
808 prop_binCRWS_muon2016M_bin6_TVX
809 prop_binCRWS_muon2016M_bin6_Triboson
810 prop_binCRWS_muon2017_bin6_Triboson
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1

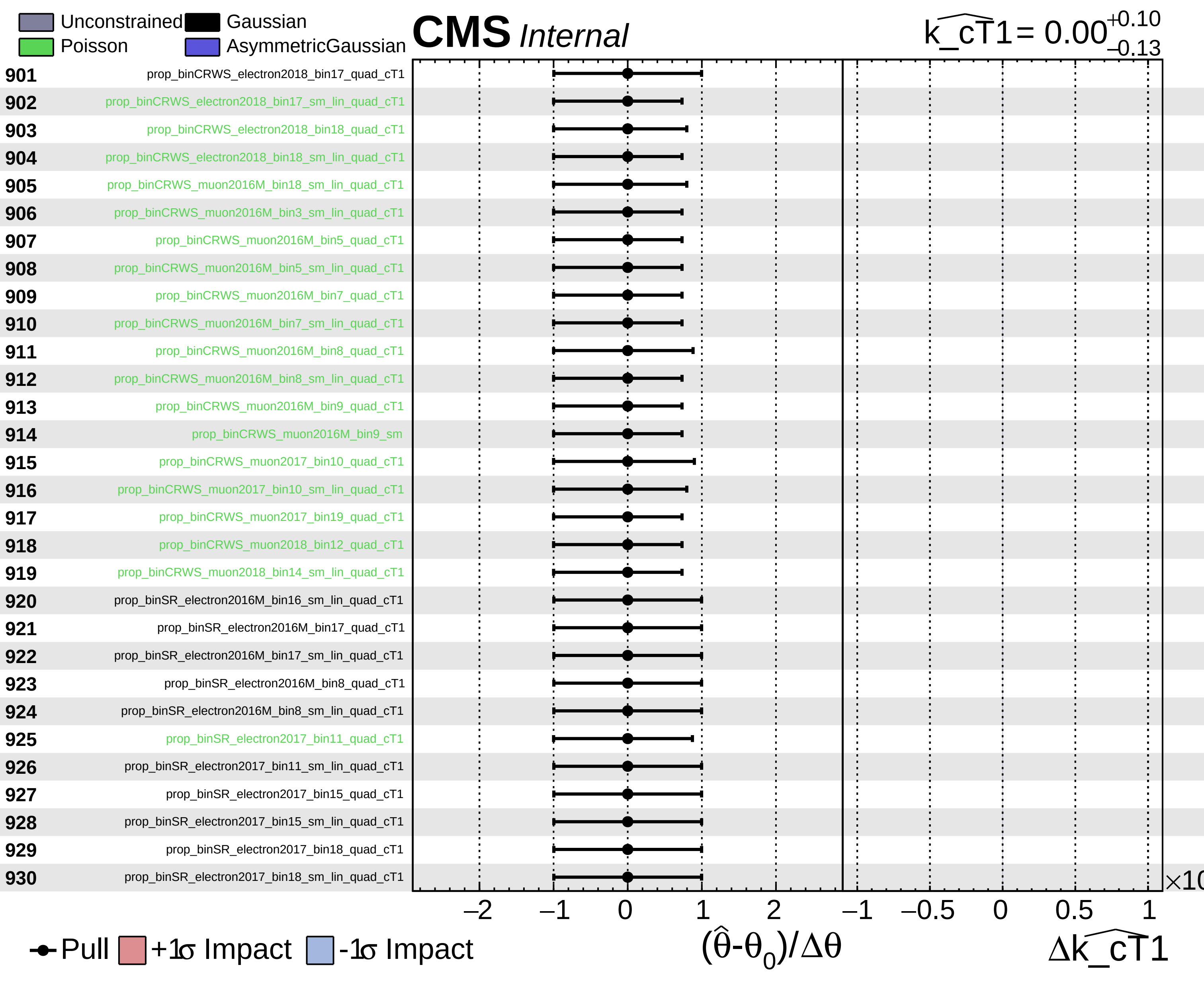

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = 0.00^{+0.10}_{-0.13}$
811 prop_binCRWS_muon2018_bin5_Triboson
812 prop_binCRWS_muon2017_bin6_TVX
813 prop_binCRWS_muon2017_bin7_TVX
814 prop_binCRWS_electron2016M_bin10_TVX
815 prop_binCRWS_electron2016M_bin10_Triboson
816 prop_binCRWS_electron2016M_bin10_ZZtoLep
817 prop_binCRWS_muon2016M_bin16_ZZtoLep
818 prop_binCRWS_muon2018_bin18_Triboson
819 prop_binCRWS_electron2018_bin13_ZZtoLep
820 prop_binCRWS_muon2018_bin10_Triboson
821 prop_binCRWS_electron2018_bin13_Triboson
822 prop_binCRWS_muon2016M_bin3_Triboson
823 prop_binCRWS_electron2017_bin16
824 prop_binCRWS_muon2016M_bin7_sm
825 prop_binCRWS_electron2018_bin7_TVX
826 prop_binCRWS_electron2018_bin7_Triboson
827 prop_binCRWS_muon2016M_bin18_TVX
828 FES
829 prop_binCRWS_electron2018_bin14_TVX
830 prop_binCRWS_electron2018_bin14_ZZtoLep
831 prop_binCRWS_electron2018_bin14_Triboson
832 prop_binCRWS_muon2017_bin8_TVX
833 prop_binCRWS_muon2017_bin5_Triboson
834 prop_binCRWS_muon2016M_bin4_Triboson
835 prop_binCRWS_muon2016M_bin9_TVX
836 prop_binCRWS_muon2016M_bin5_VG
837 prop_binCRWS_electron2016M_bin16_sm
838 prop_binCRWS_muon2016M_bin8_TVX
839 prop_binCRWS_electron2017_bin5_TVX
840 prop_binCRWS_muon2016M_bin3_VG
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



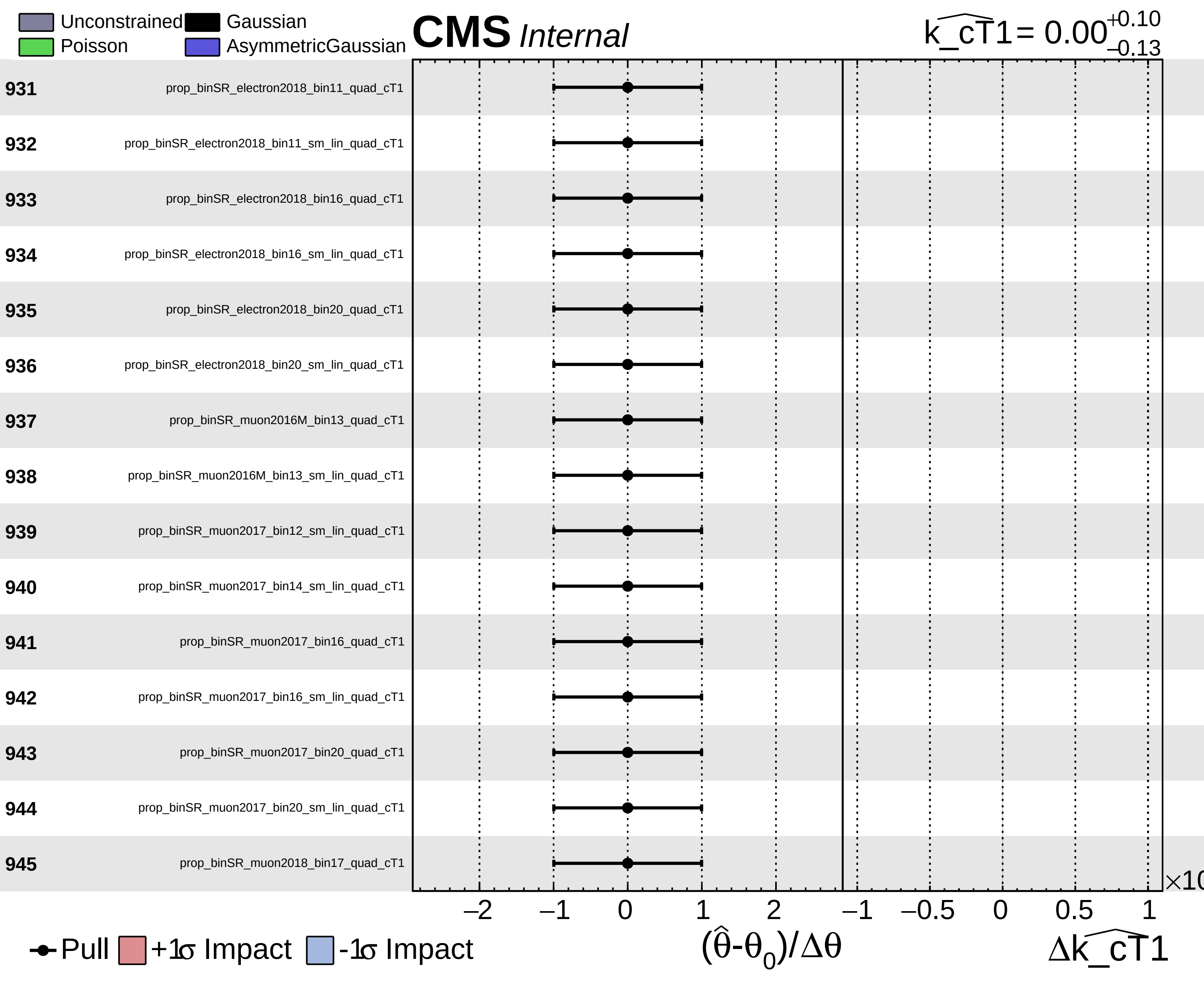
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
841 prop_binCRWS_muon2016M_bin16_TVX
842 prop_binCRWS_electron2018_bin13_TVX
843 prop_binSR_muon2016M_bin19_sm_lin_quad_cT1
844 prop_binCRWS_muon2016M_bin5_Triboson
845 prop_binCRWS_electron2016M_bin7_sm
846 prop_binSR_electron2017_bin17_sm_lin_quad_cT1
847 prop_binSR_muon2016M_bin19_quad_cT1
848 prop_binSR_electron2018_bin14_quad_cT1
849 prop_binSR_muon2016M_bin17_sm_lin_quad_cT1
850 prop_binSR_muon2018_bin17_sm_lin_quad_cT1
851 prop_binCRWS_muon2018_bin14_quad_cT1
852 prop_binSR_muon2016M_bin17_quad_cT1
853 VBS_2016M
854 VBS_2017
855 VBS_2018
856 r 1.0 +0 −0
857 tau_vsele_2016M
858 tau_vsele_2017
859 tau_vsele_2018
860 tau_vsmu_2016M
861 tau_vsmu_2017
862 tau_vsmu_2018
863 prop_binCRWS_muon2018_bin12_sm_lin_quad_cT1
864 prop_binCRWS_electron2017_bin8_quad_cT1
865 prop_binCRWS_muon2017_bin19_sm_lin_quad_cT1
866 prop_binCRWS_muon2018_bin14_TVX
867 prop_binCRWS_muon2018_bin18_sm_lin_quad_cT1
868 prop_binSR_electron2018_bin14_sm_lin_quad_cT1
869 prop_binCRWS_muon2016M_bin18_quad_cT1
870 prop_binCRWS_electron2016M_bin7_sm_lin_quad_cT1
−2 −1 0 1 2
−50 0 50
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
871 prop_binCRWS_muon2016M_bin3_quad_cT1
872 prop_binCRWS_muon2018_bin12_TVX
873 prop_binCRWS_electron2016M_bin10_sm
874 prop_binCRWS_muon2017_bin10_Triboson
875 prop_binCRWS_muon2018_bin7_Triboson
876 prop_binCRWS_electron2018_bin18_Triboson
877 prop_binCRWS_electron2016M_bin17_sm_lin_quad_cT1
878 prop_binSR_electron2016M_bin16_quad_cT1
879 prop_binSR_muon2017_bin12_quad_cT1
880 prop_binCRWS_electron2016M_bin17_Triboson
881 prop_binSR_electron2017_bin17_quad_cT1
882 prop_binCRWS_muon2017_bin13_TVX
883 prop_binSR_muon2017_bin14_quad_cT1
884 prop_binCRWS_muon2016M_bin18_sm
885 prop_binCRWS_electron2016M_bin6_WrongSign
886 prop_binCRWS_muon2018_bin12_Triboson
887 prop_binCRWS_muon2016M_bin9_sm_lin_quad_cT1
888 prop_binCRWS_electron2016M_bin16_sm_lin_quad_cT1
889 prop_binCRWS_electron2016M_bin7_quad_cT1
890 prop_binCRWS_electron2016M_bin20_quad_cT1
891 prop_binCRWS_electron2016M_bin10_sm_lin_quad_cT1
892 prop_binCRWS_muon2018_bin18_quad_cT1
893 prop_binCRWS_electron2016M_bin10_quad_cT1
894 prop_binCRWS_electron2016M_bin16_quad_cT1
895 prop_binCRWS_electron2016M_bin17_quad_cT1
896 prop_binCRWS_electron2016M_bin17_sm
897 prop_binCRWS_electron2016M_bin20_sm_lin_quad_cT1
898 prop_binCRWS_electron2017_bin17_quad_cT1
899 prop_binCRWS_electron2017_bin17_sm_lin_quad_cT1
900 prop_binCRWS_electron2017_bin8_sm_lin_quad_cT1
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
901 prop_binCRWS_electron2018_bin17_quad_cT1
902 prop_binCRWS_electron2018_bin17_sm_lin_quad_cT1
903 prop_binCRWS_electron2018_bin18_quad_cT1
904 prop_binCRWS_electron2018_bin18_sm_lin_quad_cT1
905 prop_binCRWS_muon2016M_bin18_sm_lin_quad_cT1
906 prop_binCRWS_muon2016M_bin3_sm_lin_quad_cT1
907 prop_binCRWS_muon2016M_bin5_quad_cT1
908 prop_binCRWS_muon2016M_bin5_sm_lin_quad_cT1
909 prop_binCRWS_muon2016M_bin7_quad_cT1
910 prop_binCRWS_muon2016M_bin7_sm_lin_quad_cT1
911 prop_binCRWS_muon2016M_bin8_quad_cT1
912 prop_binCRWS_muon2016M_bin8_sm_lin_quad_cT1
913 prop_binCRWS_muon2016M_bin9_quad_cT1
914 prop_binCRWS_muon2016M_bin9_sm
915 prop_binCRWS_muon2017_bin10_quad_cT1
916 prop_binCRWS_muon2017_bin10_sm_lin_quad_cT1
917 prop_binCRWS_muon2017_bin19_quad_cT1
918 prop_binCRWS_muon2018_bin12_quad_cT1
919 prop_binCRWS_muon2018_bin14_sm_lin_quad_cT1
920 prop_binSR_electron2016M_bin16_sm_lin_quad_cT1
921 prop_binSR_electron2016M_bin17_quad_cT1
922 prop_binSR_electron2016M_bin17_sm_lin_quad_cT1
923 prop_binSR_electron2016M_bin8_quad_cT1
924 prop_binSR_electron2016M_bin8_sm_lin_quad_cT1
925 prop_binSR_electron2017_bin11_quad_cT1
926 prop_binSR_electron2017_bin11_sm_lin_quad_cT1
927 prop_binSR_electron2017_bin15_quad_cT1
928 prop_binSR_electron2017_bin15_sm_lin_quad_cT1
929 prop_binSR_electron2017_bin18_quad_cT1
930 prop_binSR_electron2017_bin18_sm_lin_quad_cT1
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = 0.00 +0.10 −0.13
931 prop_binSR_electron2018_bin11_quad_cT1
932 prop_binSR_electron2018_bin11_sm_lin_quad_cT1
933 prop_binSR_electron2018_bin16_quad_cT1
934 prop_binSR_electron2018_bin16_sm_lin_quad_cT1
935 prop_binSR_electron2018_bin20_quad_cT1
936 prop_binSR_electron2018_bin20_sm_lin_quad_cT1
937 prop_binSR_muon2016M_bin13_quad_cT1
938 prop_binSR_muon2016M_bin13_sm_lin_quad_cT1
939 prop_binSR_muon2017_bin12_sm_lin_quad_cT1
940 prop_binSR_muon2017_bin14_sm_lin_quad_cT1
941 prop_binSR_muon2017_bin16_quad_cT1
942 prop_binSR_muon2017_bin16_sm_lin_quad_cT1
943 prop_binSR_muon2017_bin20_quad_cT1
944 prop_binSR_muon2017_bin20_sm_lin_quad_cT1
945 prop_binSR_muon2018_bin17_quad_cT1
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1